



Histopathology

Cell injury

rd class3

2 Lab

2026-2027

FIBMS. Pathology
Dr.Duha Talib Kareem

M.Sc. Pathology
Dr.ahmed A. Hussein

M.Sc. pathology
Dr. Mahdi Jasim

Learning objects

,Define cell injury and list the primary patterns of tissue necrosis (coagulative.1
. (liquefactive, caseous, gangrenous, fat, and fibrinoid necrosis

,Explain the fundamental differences between reversible and irreversible cell injury .2
.and contrast the characteristics of necrosis versus apoptosis

,Describe how tissues undergo cellular adaptations—such as hypertrophy .3
hyperplasia, atrophy, and metaplasia—in response to environmental stress or
.changing functional demands

Examine the mechanisms and morphological features of intracellular .4
accumulations (such as fatty change or steatosis) and endogenous/exogenous
. (pigments (such as lipofuscin, bilirubin, and hemosiderin

Differentiate between dystrophic calcification and metastatic calcification based .5
on the condition of the affected tissue (damaged/dead versus normal) and the
.underlying disease states



Cell injury

Cell injury

Cells actively interact with their environment, adjusting their structure and •
.function to accommodate changing demands and extracellular stresses

The intracellular environment of cells is tightly regulated such that it •
.remains fairly constant, a state referred to as **homeostasis**

(As cells encounter physiologic stresses (increased workload in the heart •
or potentially injurious conditions (nutrient deprivation), they can undergo
adaptation, achieving a new steady state and preserving viability and
.function

If the adaptive capability is exceeded or if the external stress is inherently •
.harmful or excessive, **cell injury** develops



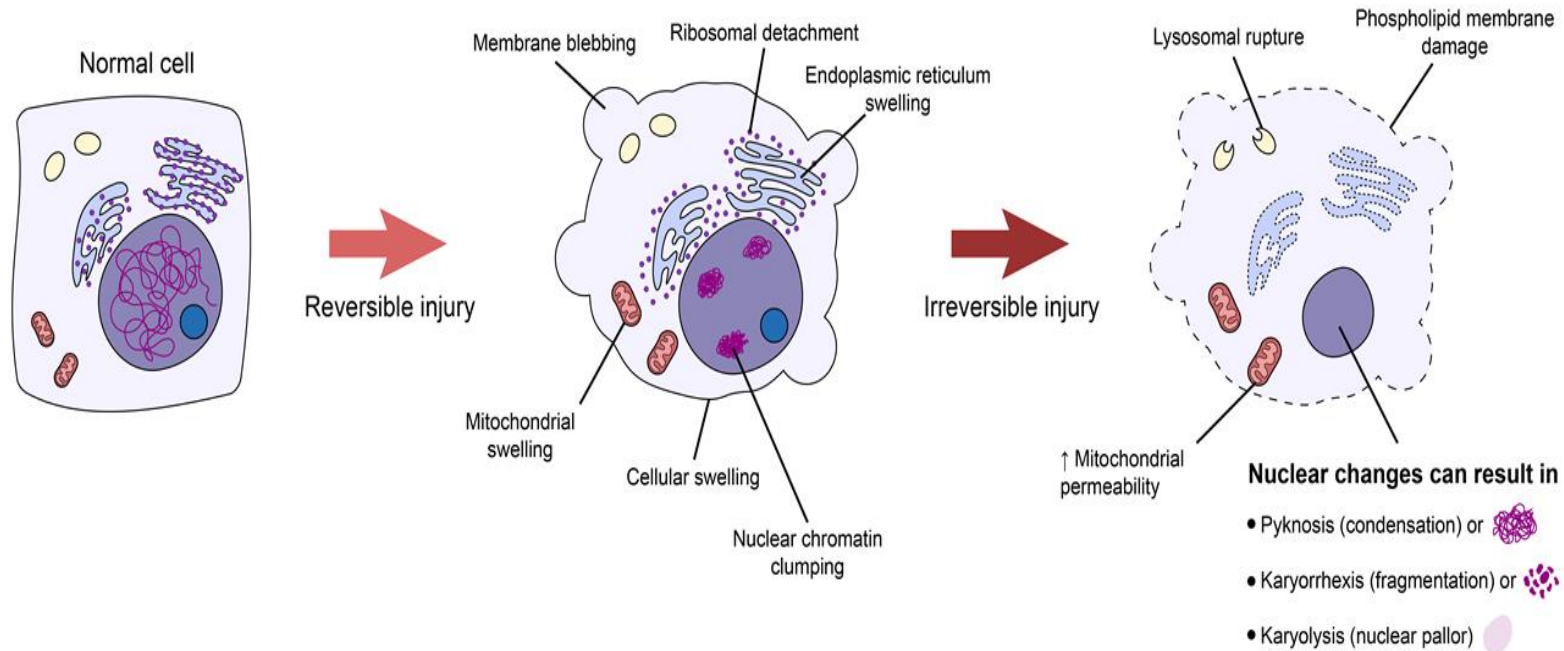
Within certain limits, injury is **reversible**, and cells return to their stable baseline; however, if the stress is severe, persistent, or rapid in onset, it results in **irreversible injury** and death of the affected cells

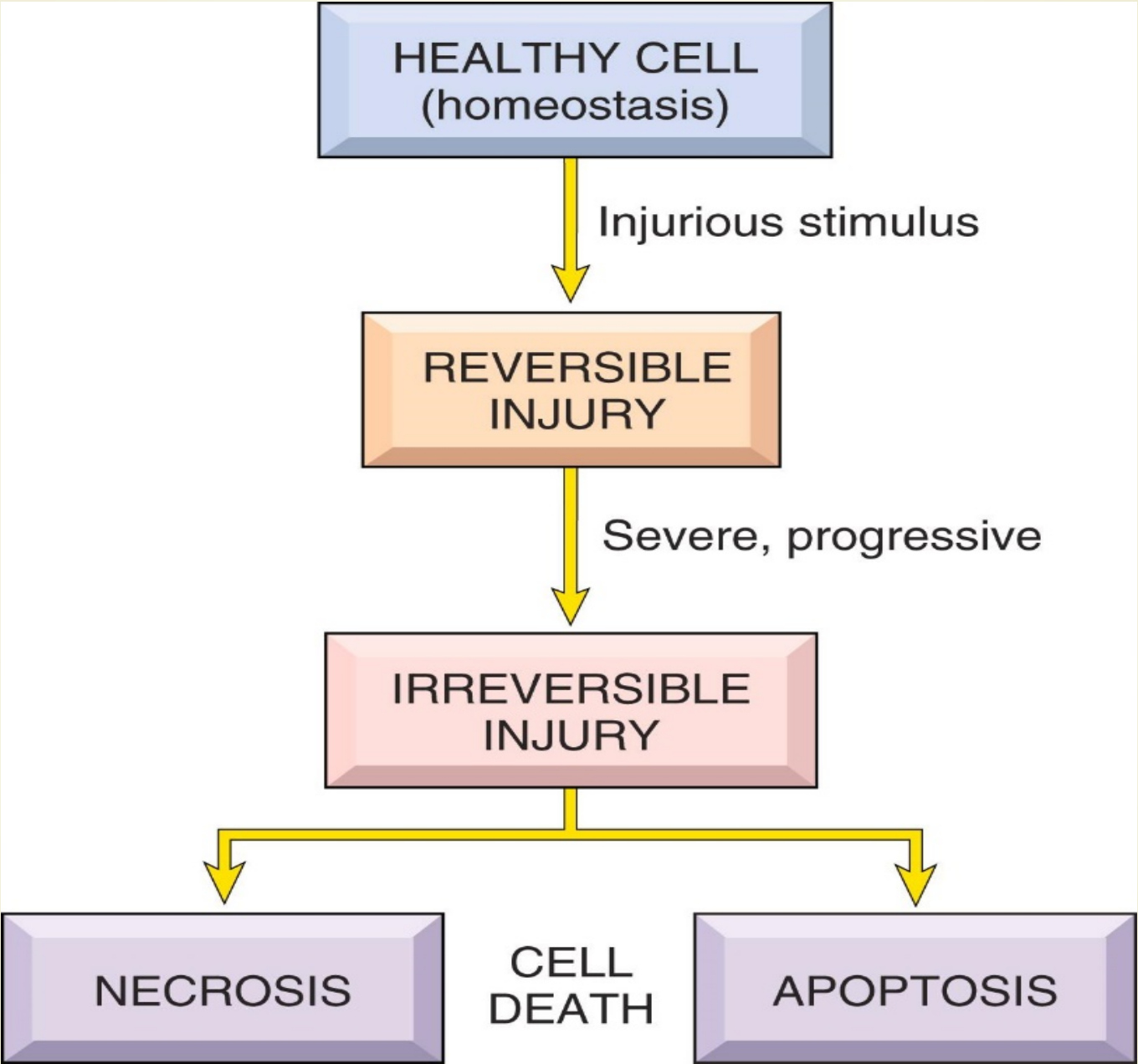
Cell death is one of the most crucial events in the evolution of disease •
in any tissue or organ

Cell death results from diverse causes, including ischemia (lack of blood •
flow), infections, toxins, and immune reactions

Cell death also is a normal and essential process in embryogenesis, the •
The development of organs, and the maintenance of tissue homeostasis

Cell Injury





```
graph TD; A[HEALTHY CELL (homeostasis)] -- "Injurious stimulus" --> B[REVERSIBLE INJURY]; B -- "Severe, progressive" --> C[IRREVERSIBLE INJURY]; C --> D[NECROSIS]; C --> E[APOPTOSIS]; D --- F[CELL DEATH]; E --- F
```

HEALTHY CELL
(homeostasis)

Injurious stimulus

REVERSIBLE
INJURY

Severe, progressive

IRREVERSIBLE
INJURY

NECROSIS

CELL
DEATH

APOPTOSIS

Apoptosis

is a process that eliminates cells with a variety of intrinsic abnormalities and •
promotes clearance of the fragments of the dead cells without eliciting an
.inflammatory reaction

Unlike necrosis, which is always an indication of a pathologic process, apoptosis •
.also occurs in healthy tissues

:It serves to •

eliminate unwanted cells during normal development.1

maintain constant cell numbers, so it is not necessarily associated with pathologic .2
.cell injury

.These types of **physiologic cell death** are also called **programmed cell death** •

Necrosis

Necrosis is a form of **cell death** in which cellular membranes fall •
.apart, and cellular enzymes leak out and ultimately digest the cell

.Necrosis elicits a local host reaction, called inflammation •

.Inflammation is induced by substances released from dead cells •

Inflammation serves to eliminate the debris and start the repair •
.process

Morphologic features of necrosis

Cytoplasmic changes. the necrotic cells show increased cytoplasmic -1
.eosinophilia i.e. it appears deep pink in color than normal cells

(**Nuclear changes** (Pyknosis, Karyorrhexis, Karyolysis -2

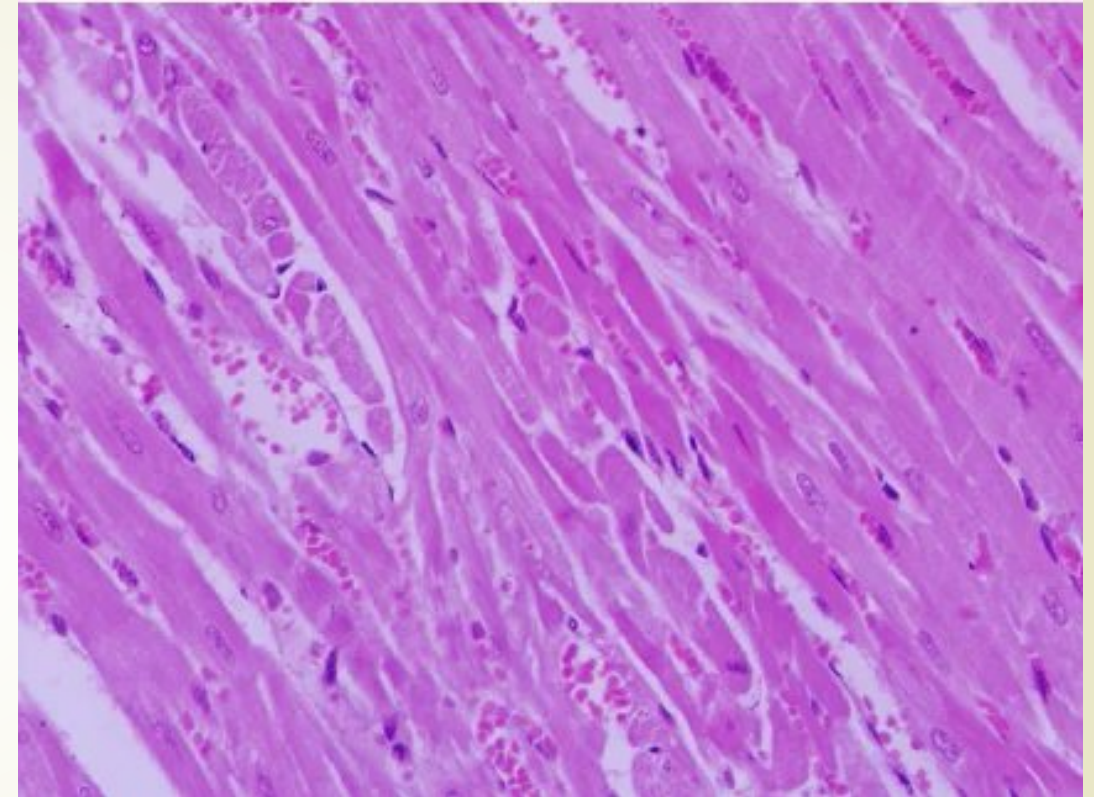
Patterns of tissue necrosis

Type	Mechanism	Pathologic Changes
Coagulative necrosis	Most often results from interruption of blood supply, resulting in denaturation of proteins; best seen in organs supplied by end arteries with limited collateral circulation, such as the heart and kidney	General architecture well preserved, except for nuclear changes; increased cytoplasmic binding of acidophilic dyes
Liquefactive necrosis	Enzymatic liquefaction of necrotic tissue, most often in the CNS, where it is caused by interruption of blood supply; also occurs in areas of bacterial infection	Necrotic tissue soft and liquefied
Caseous necrosis	Shares features of both coagulation and liquefaction necrosis; most commonly seen in tuberculous granulomas	Architecture not preserved but tissue not liquefied; gross appearance is soft and cheese-like; histologic appearance is amorphous, with increased affinity for acidophilic dyes
Gangrenous necrosis	Most often results from interruption of blood supply to a lower extremity or the bowel	Changes depend on tissue involved and whether gangrene is dry or wet
Fibrinoid necrosis	Characterized by deposition of fibrin-like proteinaceous material in walls of arteries; often observed as part of immune-mediated vasculitis	Smudgy pink appearance in vascular wall; actual necrosis may or may not be present
Fat necrosis	Liberation of pancreatic enzymes with autodigestion of pancreatic parenchyma; trauma to fat cells	Necrotic fat cells, acute inflammation, hemorrhage, calcium soap formation, clustering of lipid-laden macrophages (in the pancreas)

Coagulative necrosis

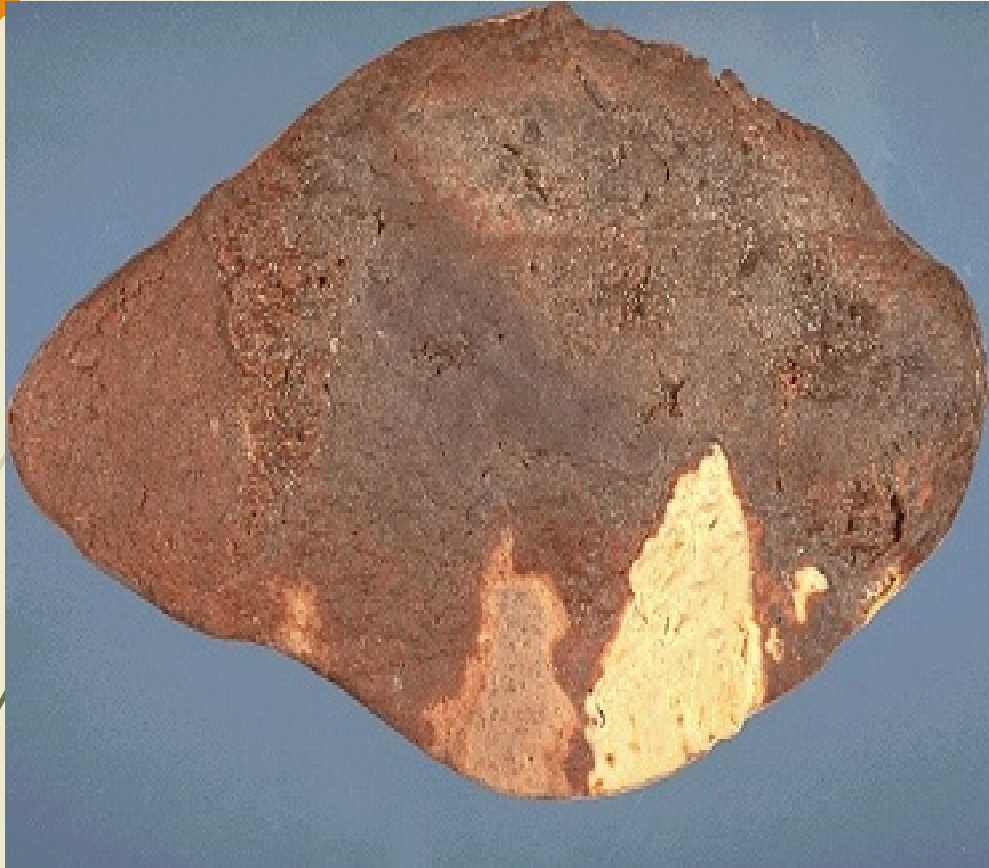


A and B, On gross examination, 2 lesions involving the myocardium and the papillary muscles are seen. Green arrows point to dark mottling, and the black arrows point to a yellow, softened lesion with red-tan borders; these correspond to a myocardial infarction in between 12 to 24 hours and 10 to 14 days, respectively.



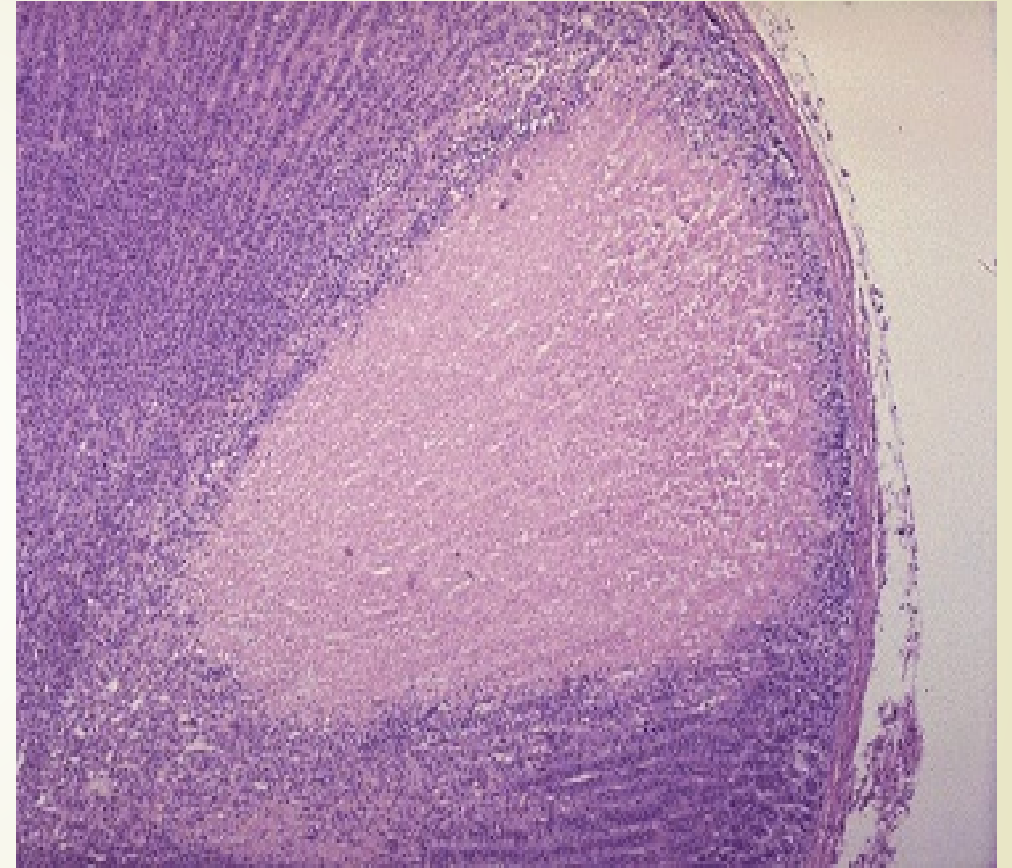
Coagulation necrosis of the heart muscle fibers distinct eosinophilic cytoplasm, lack of transverse striations and pyknotic or absent nuclei (HE, 40x).

Coagulative necrosis, splenic infarctions, gross



Two large infarctions (areas of coagulative necrosis) are seen in this sectioned spleen. the etiology of coagulative .necrosis is usually vascular with loss of blood supply infarcts are often wedge-shaped with a base on the organ .capsule

Coagulative necrosis, adrenal infarction, microscopic

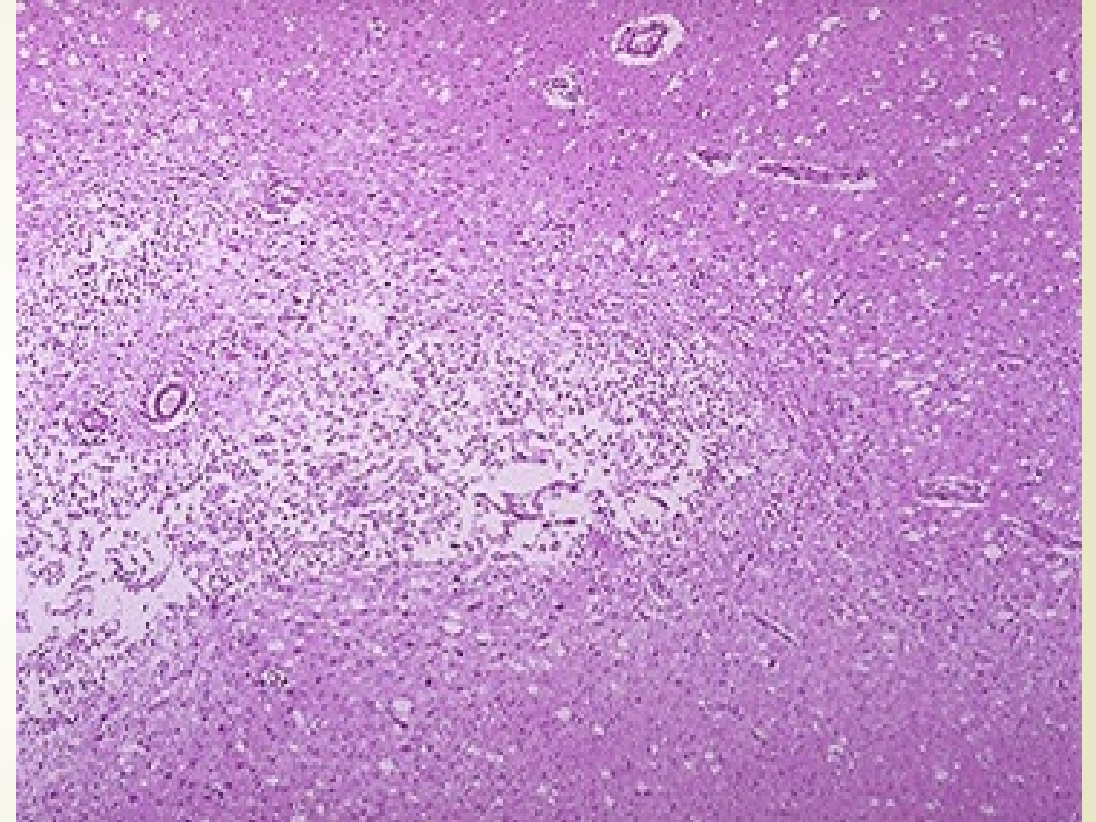


The area just under the capsule is spared because of blood supply from capsular arterial branches. This is an odd place for an infarct, but it illustrates the shape and appearance of an ischemic (pale) infarct well.

Liquefactive necrosis

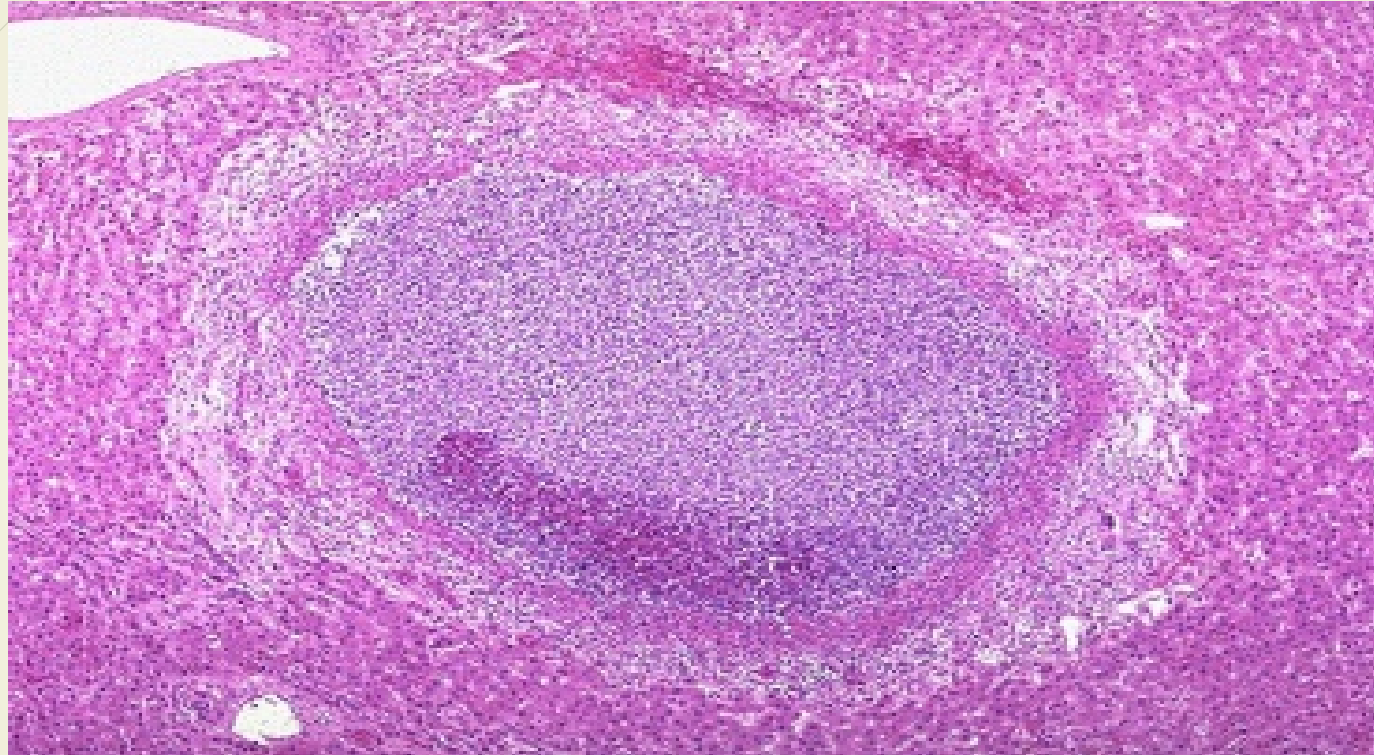


Cerebral infarction causes liquefactive necrosis due to hypoxia, and at this time, melted cells can be observed in the area as yellowish mucus masses.



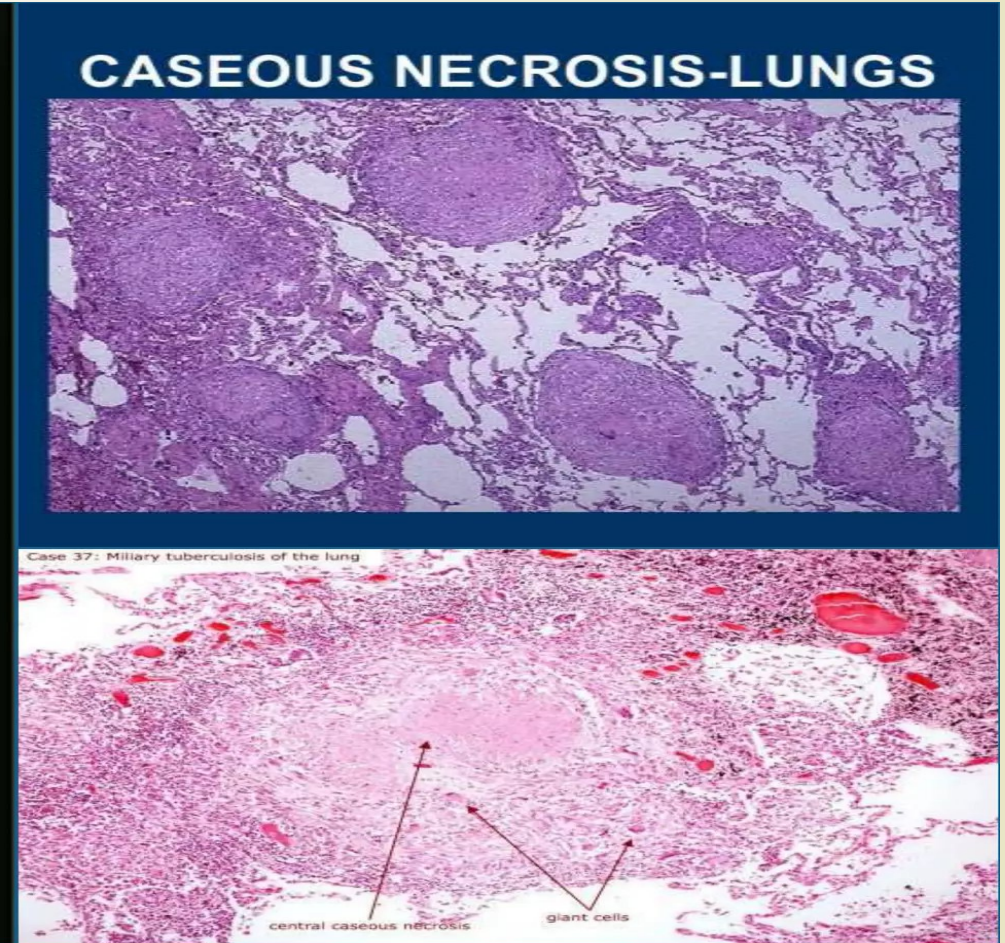
Resolution of the liquefactive necrosis by an influx of blood monocytes to become tissue macrophages in a cerebral infarction over weeks leads to the formation of a cystic space.

,Liquefactive necrosis, liver abscess microscopic



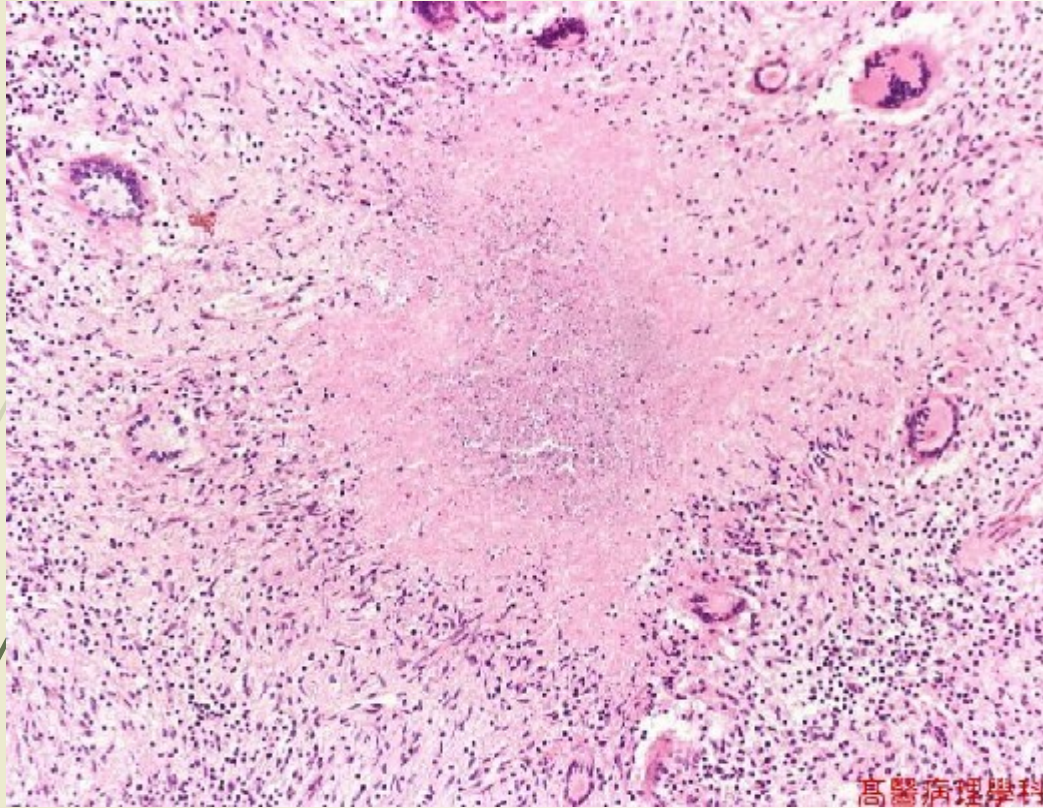
The liver shows a small abscess here filled with many neutrophils. This abscess is an example of localized liquefactive necrosis. Grossly, such an abscess appears yellow to tan because it is filled with pus (purulent exudate).

Caseous necrosis

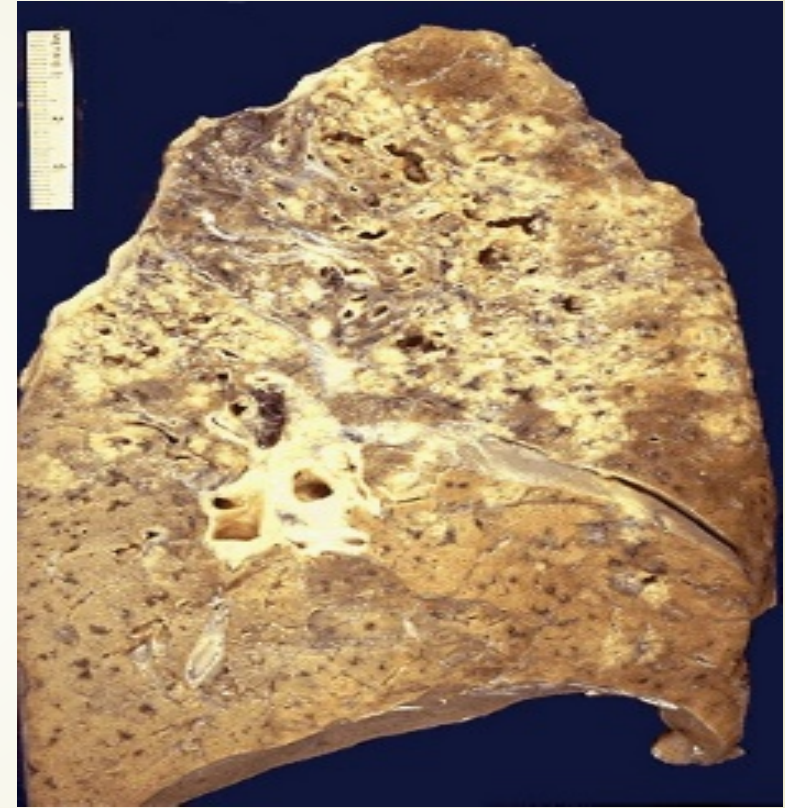


the lung containing caseous necrosis due to tuberculosis. Notice the yellow-white and cheesy debris.

Caseous necrosis of the lung



It is characterized by an aggregation or accumulation of macrophages or inflammatory cells, known as a granuloma, or tubercles.



This is more extensive caseous necrosis, with confluent cheesy tan granulomas in the upper portion of this lung in a patient with tuberculosis.

Gangrenous necrosis



Dry gangrene (gross appearance of two cases). Atherosclerosis-induced dry gangrene is seen in the foot (left).

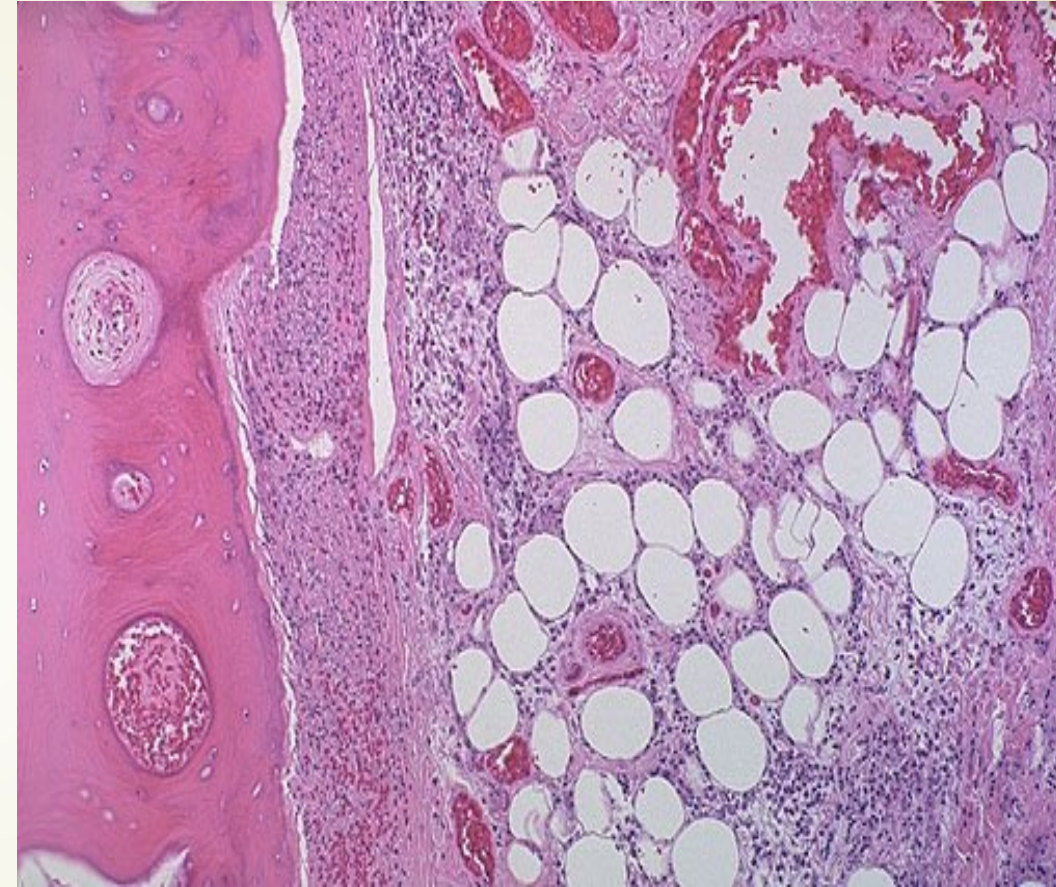


Wet gangrene (gross appearance of two cases and H&E). Infected deep irregular ulcers are formed in the back of the foot (left) and the base of the second toe after auto amputation (right).

Gangrenous necrosis

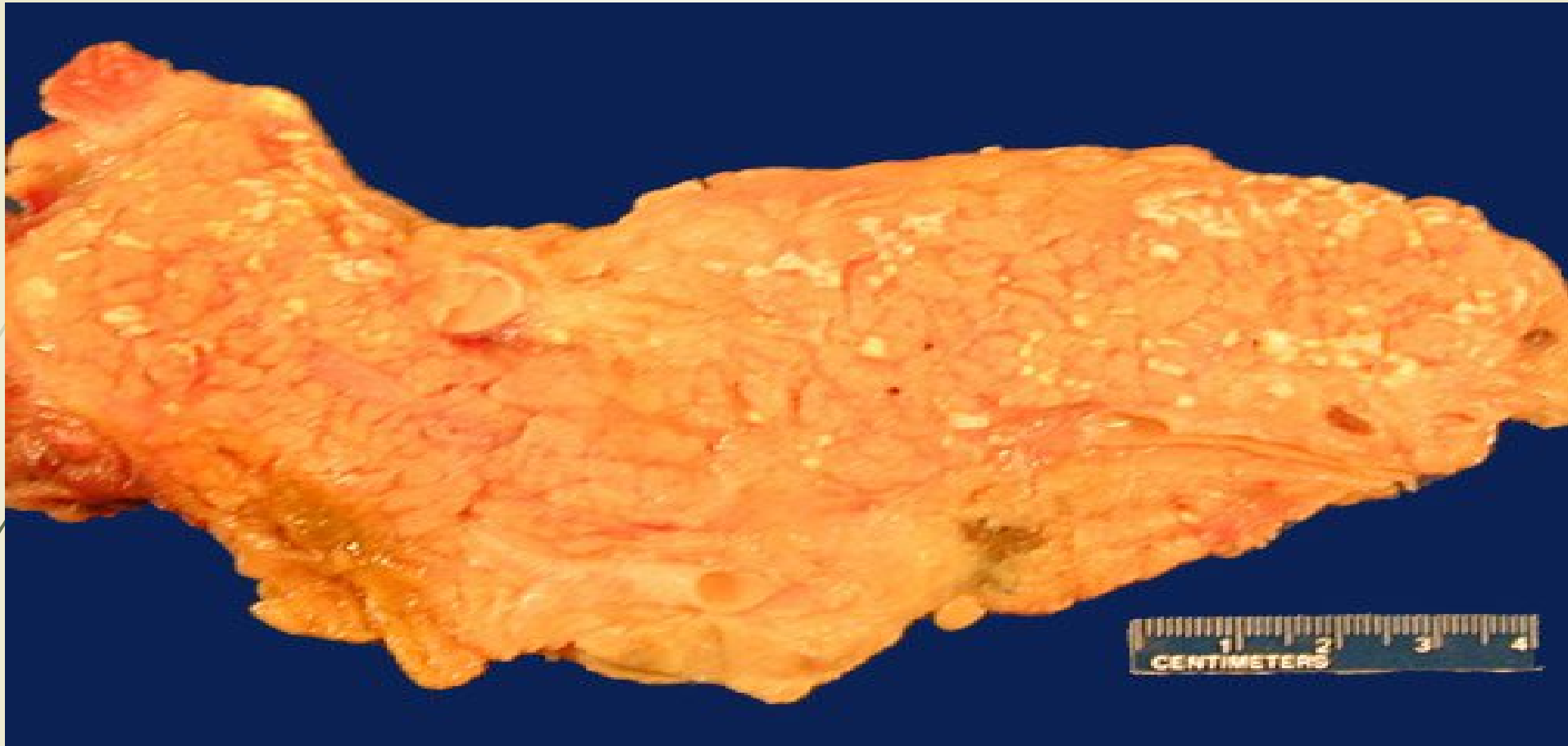


Wet gangrene there is liquefactive component from superimposed infection in addition to the coagulative necrosis from loss of blood supply. This patient had diabetes mellitus with severe peripheral vascular disease.



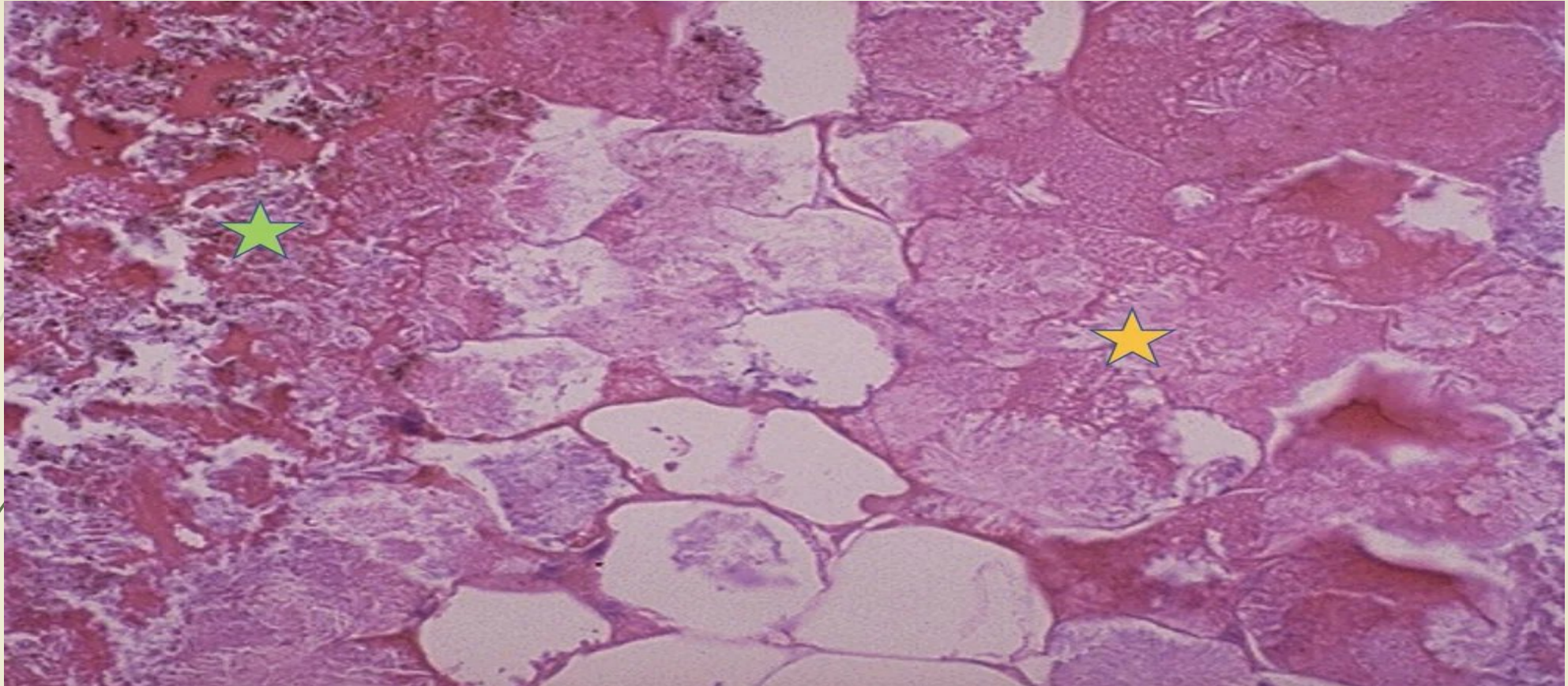
Gangrenous necrosis involves the tissues of a body part. The inflammation seen here is extending beneath the skin of a toe to involve soft tissue (fat and connective tissue at the right) and bone (at the left).

Fat necrosis



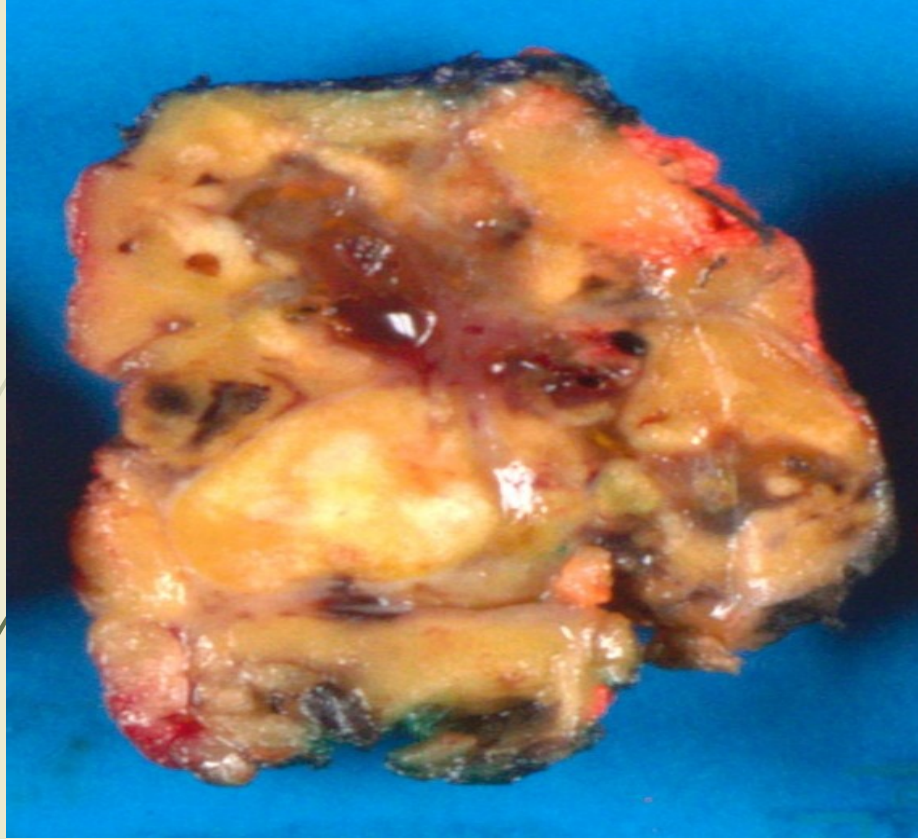
Cellular injury to the pancreatic acini leads to release of powerful enzymes which damage fat by the production of soaps, and these appear grossly as the soft, chalky white areas

Fat necrosis

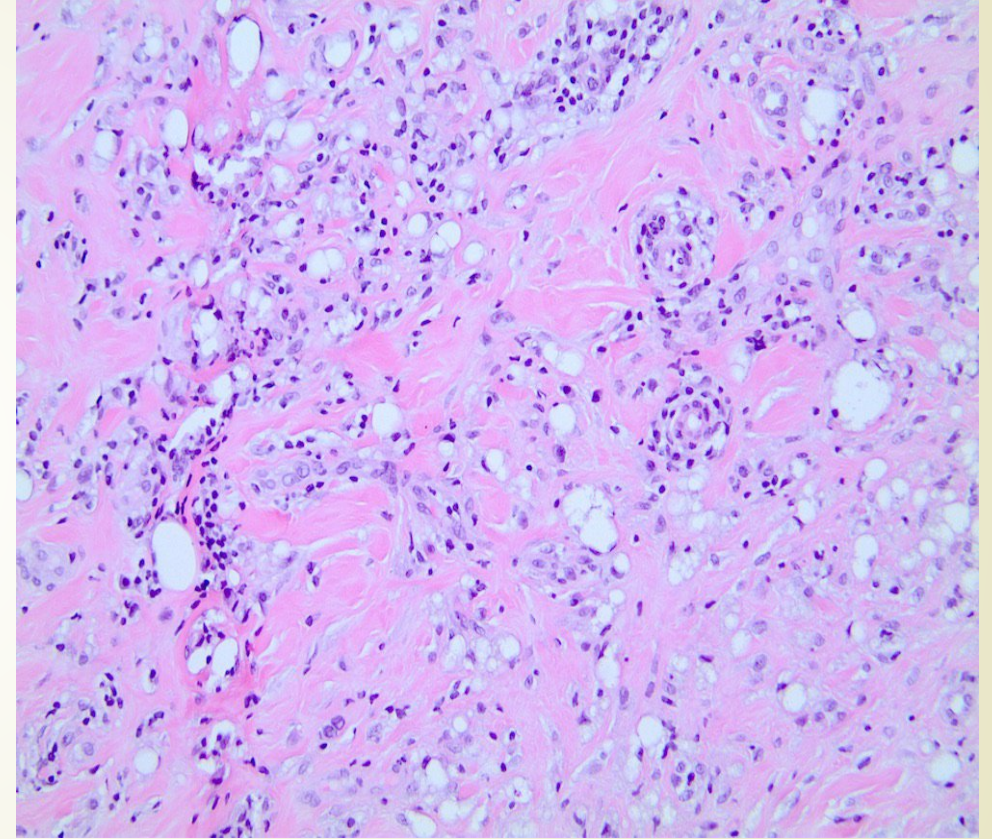


Microscopically, fat necrosis adjacent to the pancreas is seen here. There are some remaining lipocytes at the left which are not necrotic, as seen as a green star. The necrotic fat cells at the right have vague cellular outlines, have lost their peripheral nuclei, and their cytoplasm has become a pink amorphous mass of necrotic material, as seen as a yellow star.

Fat necrosis

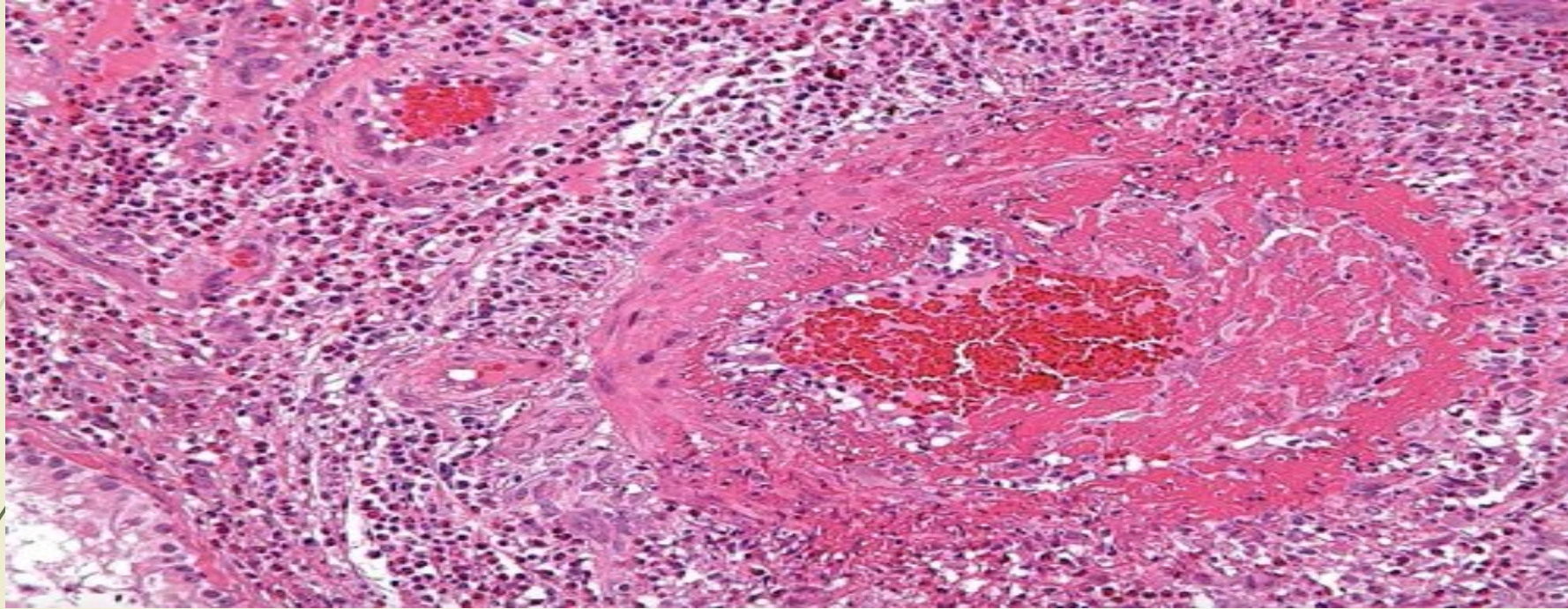


Fat necrosis within the prior biopsy site on a lumpectomy specimen.



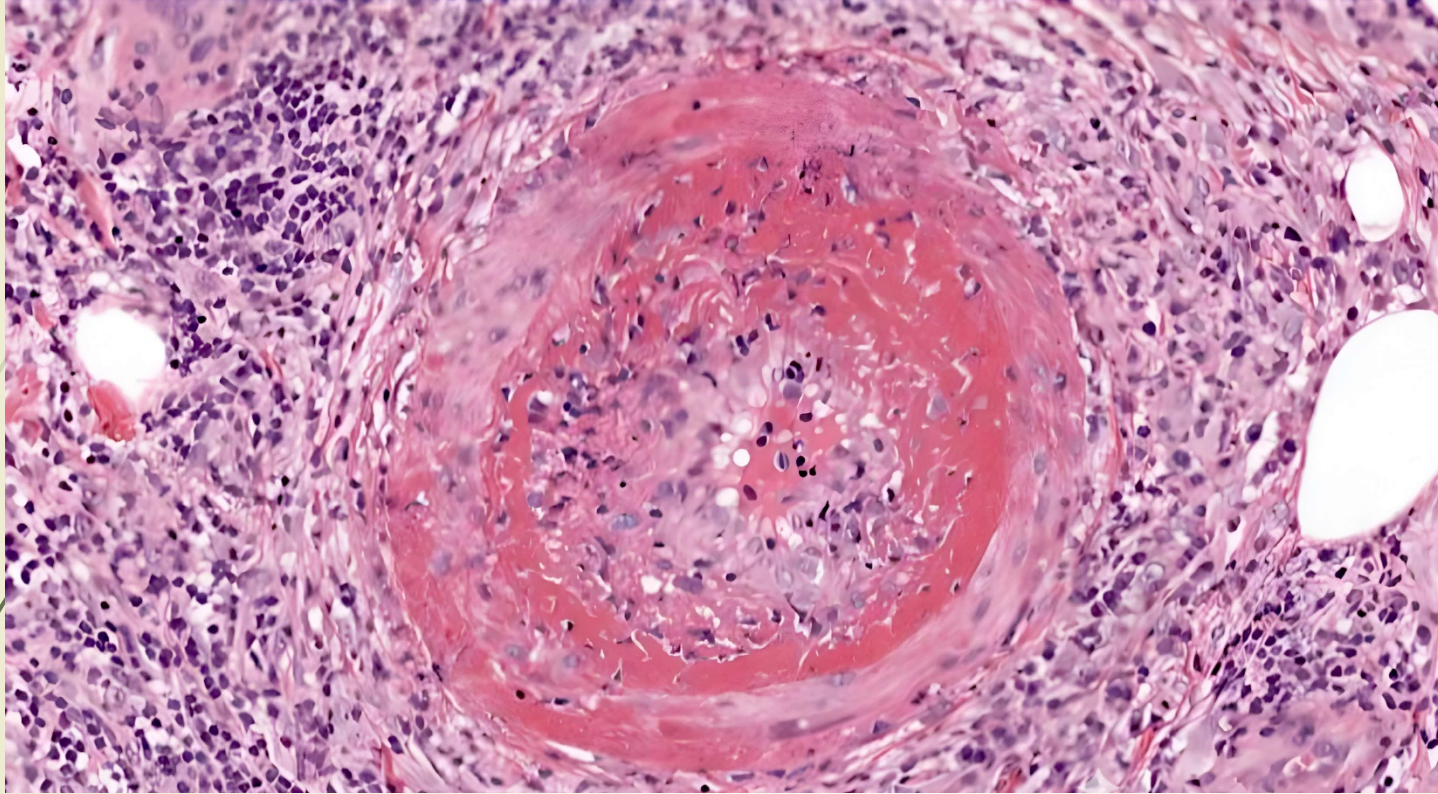
Disruption of adipocytes in early stage of fat necrosis shows fine vacuolization, diminished size and dropout of adipocytes. Mixed inflammatory cell infiltrate composed of lymphocytes, plasma cells

Fibrinoid necrosis



Microscopic picture showing (intensely pink) fibrinoid necrosis (large blood vessel - right of image) in a case of vasculitis. H&E stain was used.

Fibrinoid necrosis

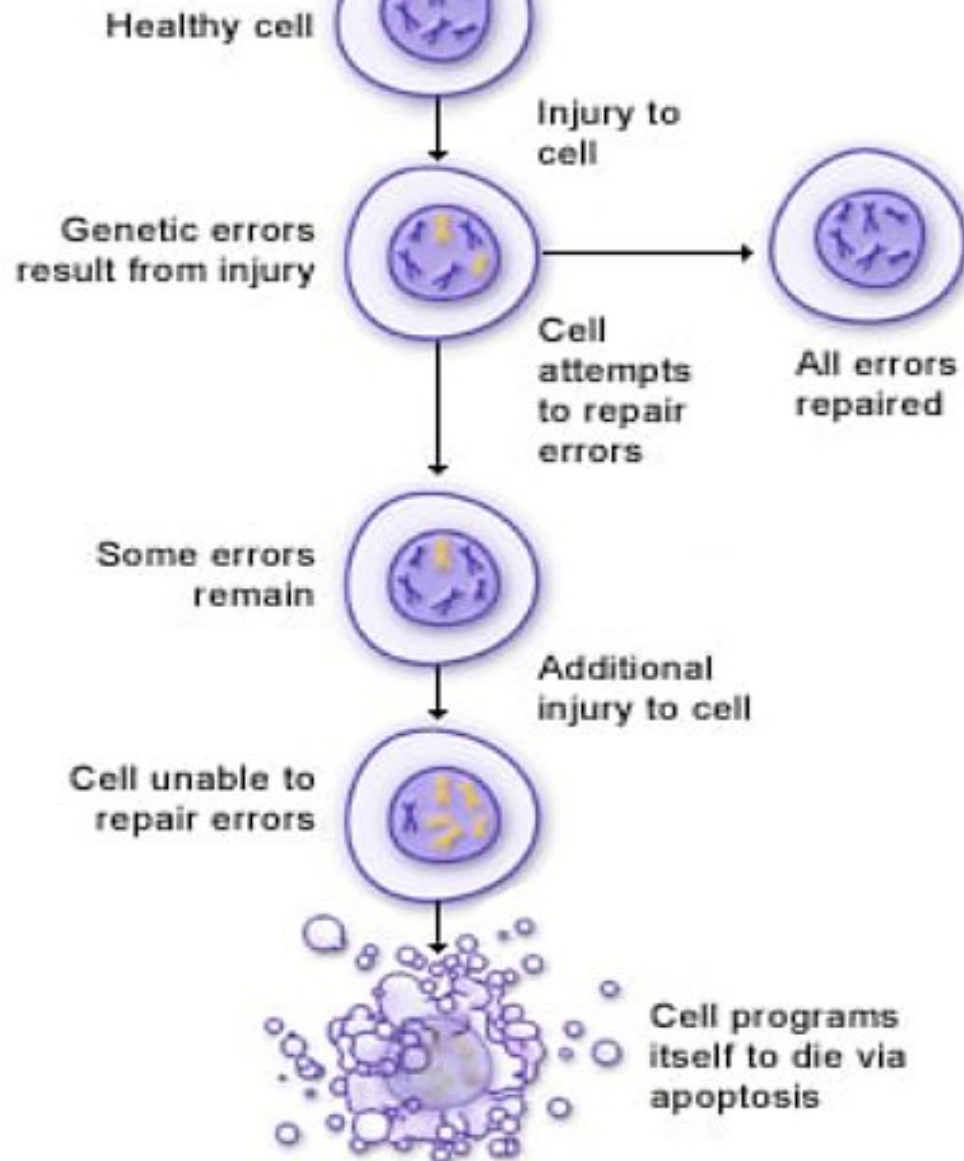


Polyarteritis nodosa showing fibrinoid necrosis of the vessel wall with intimal proliferation,

APOPTOSIS

- ❏ This form of cell death is a regulated suicide program.
- ❏ The dead cell is rapidly cleared before its contents have leaked out.
- ❏ cell death by this pathway does not elicit an inflammatory reaction in the host.

Apoptosis



- Physiologic .
- Pathologic .

Morphologic appearance of apoptotic cells

Apoptotic cells (some indicated by -
arrows) in colonic epithelium

the fragmented nuclei with -
condensed chromatin and the
shrunk cell bodies, some with
pieces falling off

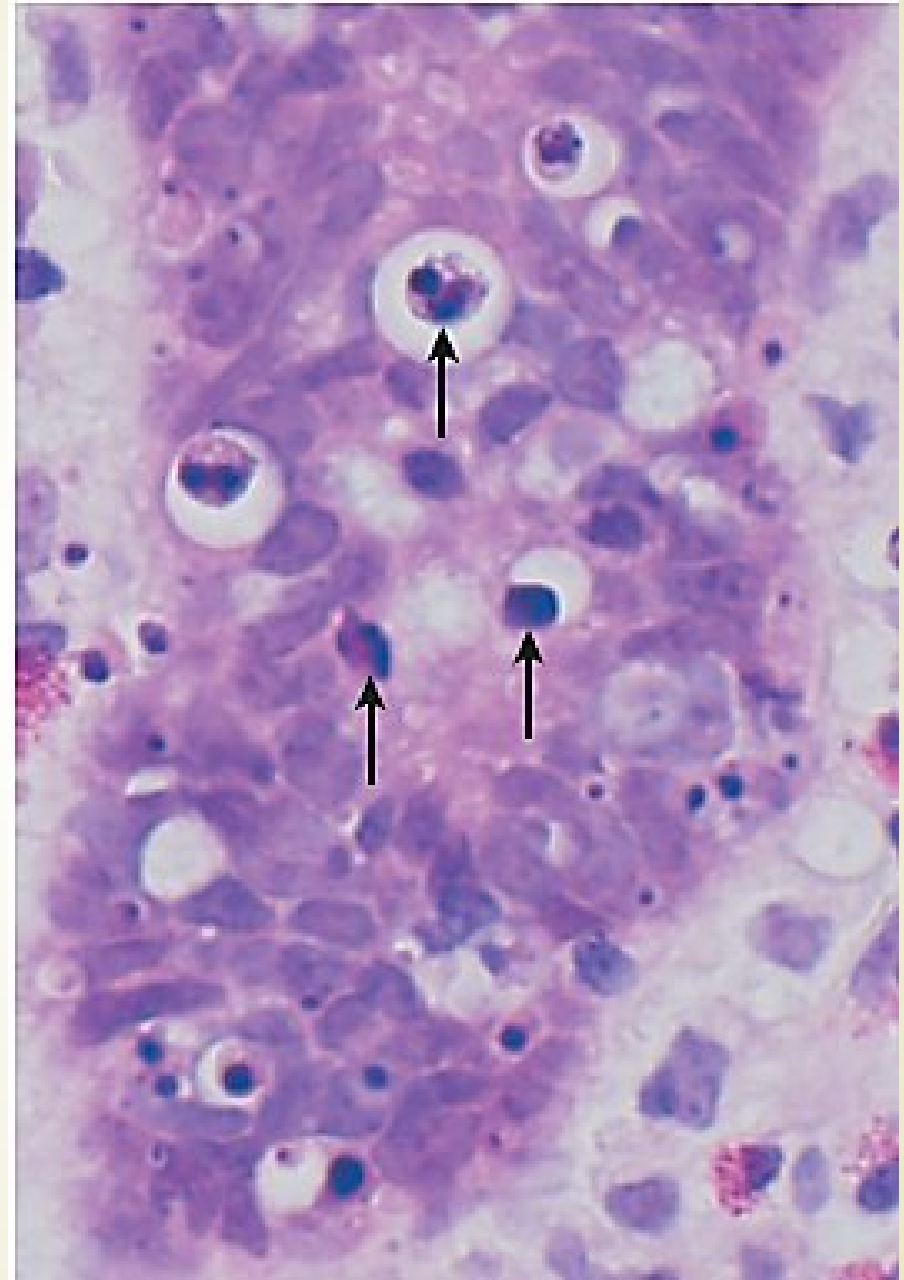


Table 2.1 Features of Necrosis and Apoptosis

Feature	Necrosis	Apoptosis
Cell size	Enlarged (swelling)	Reduced (shrinkage)
Nucleus	Pyknosis → karyorrhexis → karyolysis	Fragmentation into nucleosome-sized fragments
Plasma membrane	Disrupted	Intact; altered structure, especially orientation of lipids
Cellular contents	Enzymatic digestion; may leak out of cell	Intact; may be released in apoptotic bodies
Adjacent inflammation	Frequent	No
Physiologic or pathologic role	Invariably pathologic (culmination of irreversible cell injury)	Often physiologic means of eliminating unwanted cells; may be pathologic after some forms of cell injury, especially DNA and protein damage



CELLULAR ADAPTATIONS TO STRESS

Adaptations:

- are reversible changes in the number, size, phenotype, metabolic activity, or functions of cells in response to changes in their environment.

I. Physiologic adaptations

II. Pathologic adaptations



Hypertrophy

Hypertrophy is an increase in the size of cells resulting in increase in the size of the organ.

In pure hypertrophy: there are no new cells, just bigger cells containing increased amounts of structural proteins and organelles.
Can be physiological or pathological.

On the left is a normal uterus showing the normal mass of smooth muscle in its wall.

On the right is a uterus from a pregnant woman, in which the striking increase in mass of smooth muscle is evident.

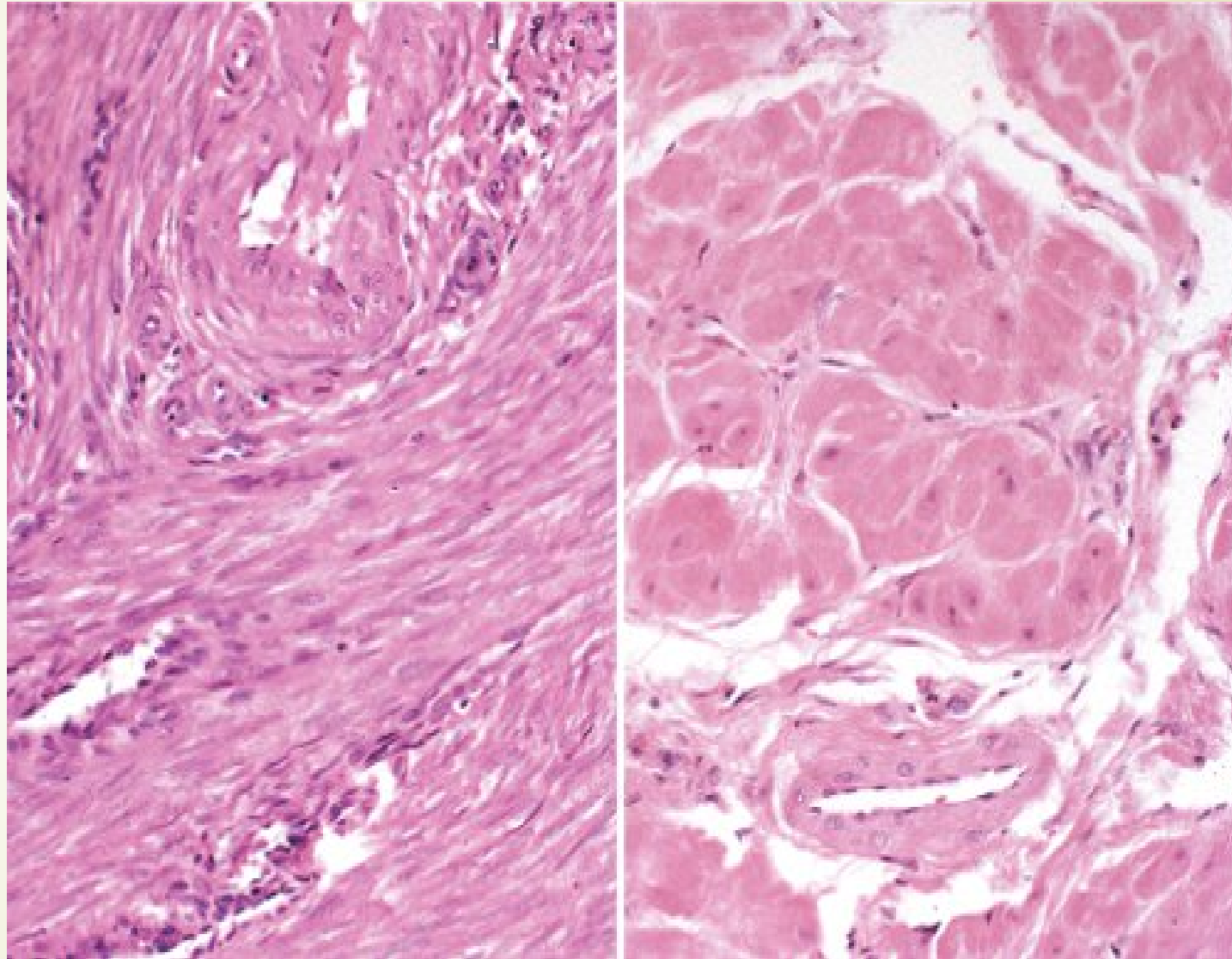
At cellular level this is due to both hyperplasia and hypertrophy of uterine smooth muscle.



uterus

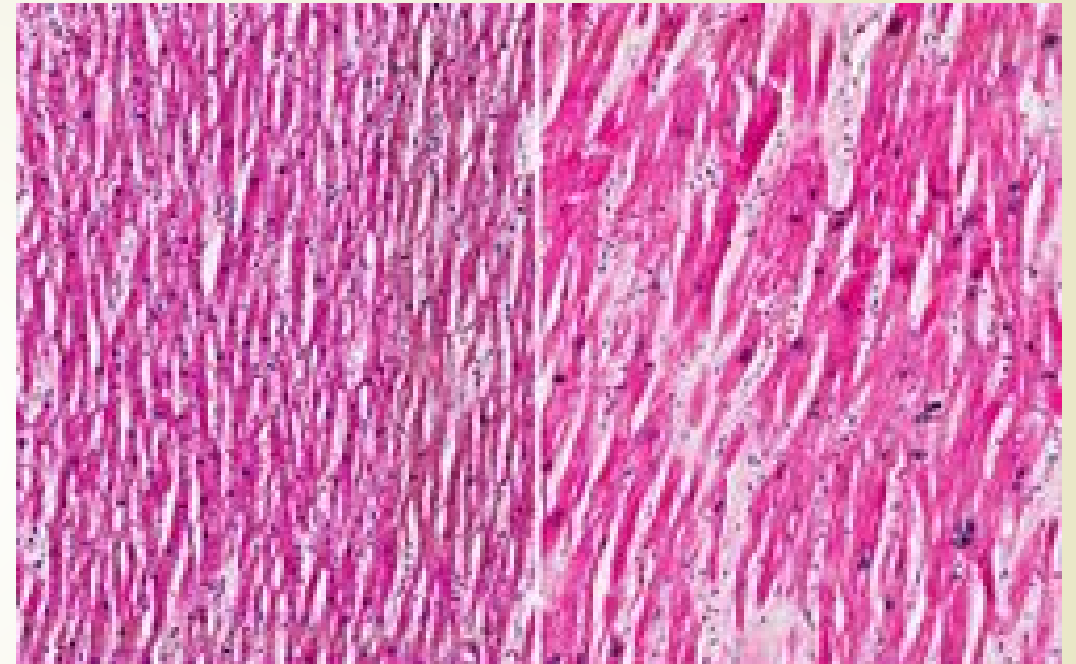
Lt. small spindle-shaped uterine smooth muscle cells from a normal uterus.

(Rt) large, plump hypertrophied smooth muscle cells from a gravid uterus (same magnification).





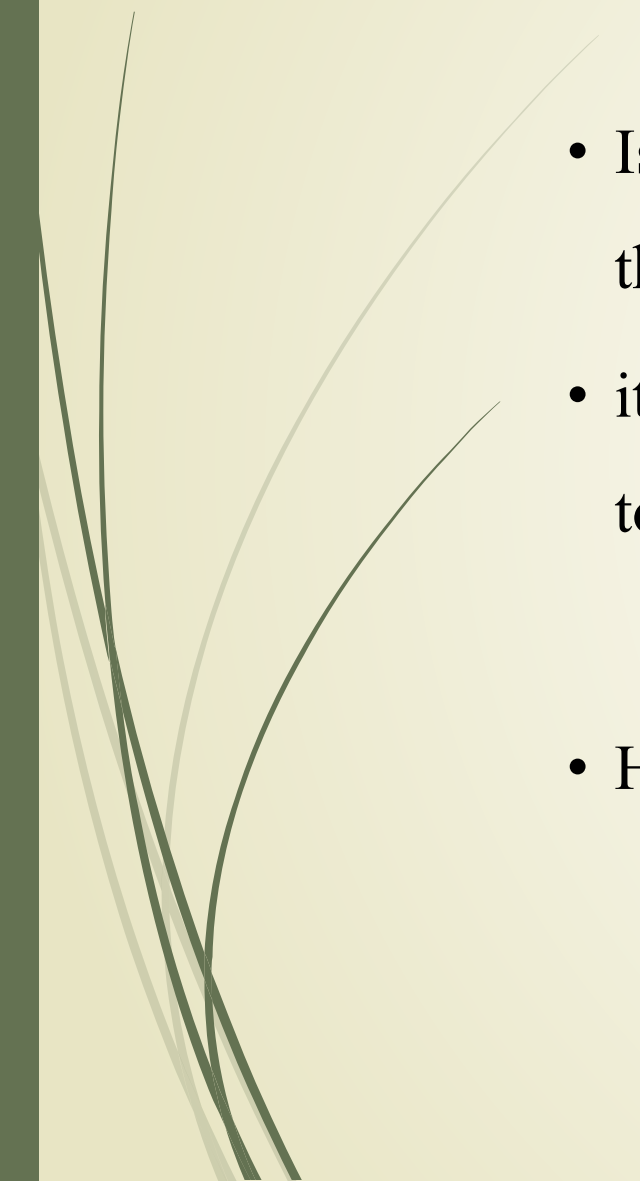
marked thickening of the left ventricle in this - patient with systemic hypertension. Not increase the size of myocardial fibers with no .increase in number



Normal myocardium (left) is compared here to hypertrophied myocardium (right). The muscle fibers are thicker and the nuclei are larger and darker in the hypertrophied myocardium.



Hyperplasia

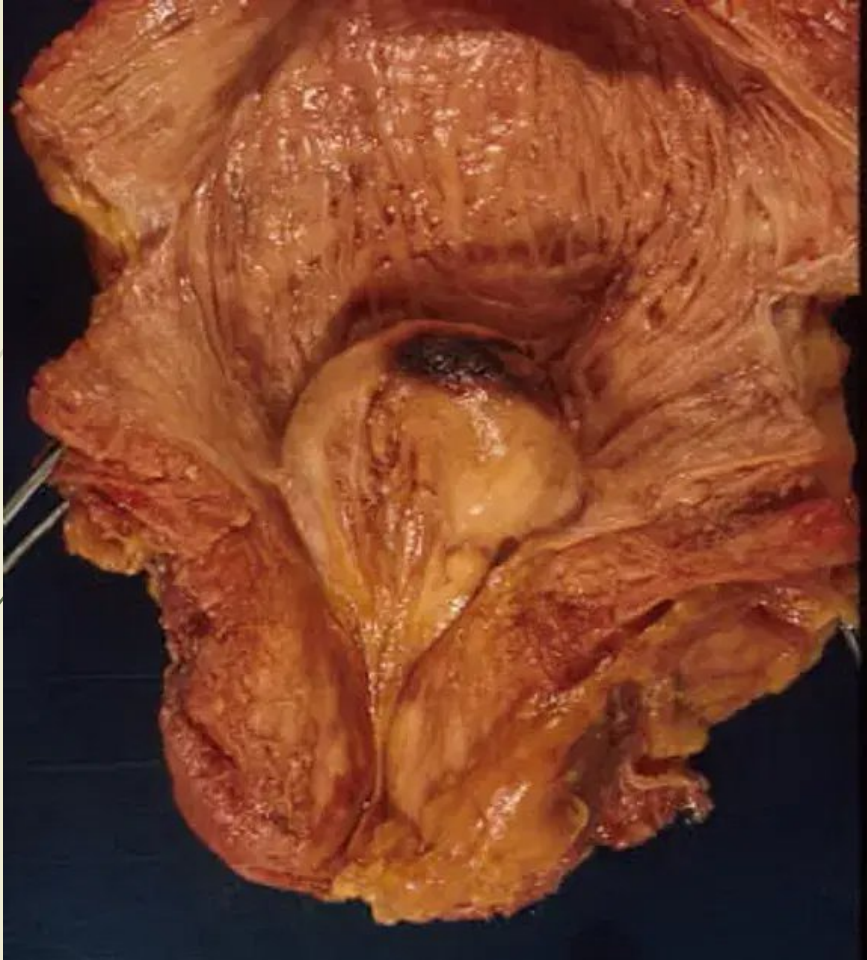
- Is an increase in the number of cells, hyperplasia takes place if the tissue contains cell populations capable of replication.
 - it may occur concurrently with hypertrophy and often in response to the same stimuli.
 - Hyperplasia can be physiologic or pathologic.
- 

The prominent folds of endometrium in this uterus are an example of .hyperplasia

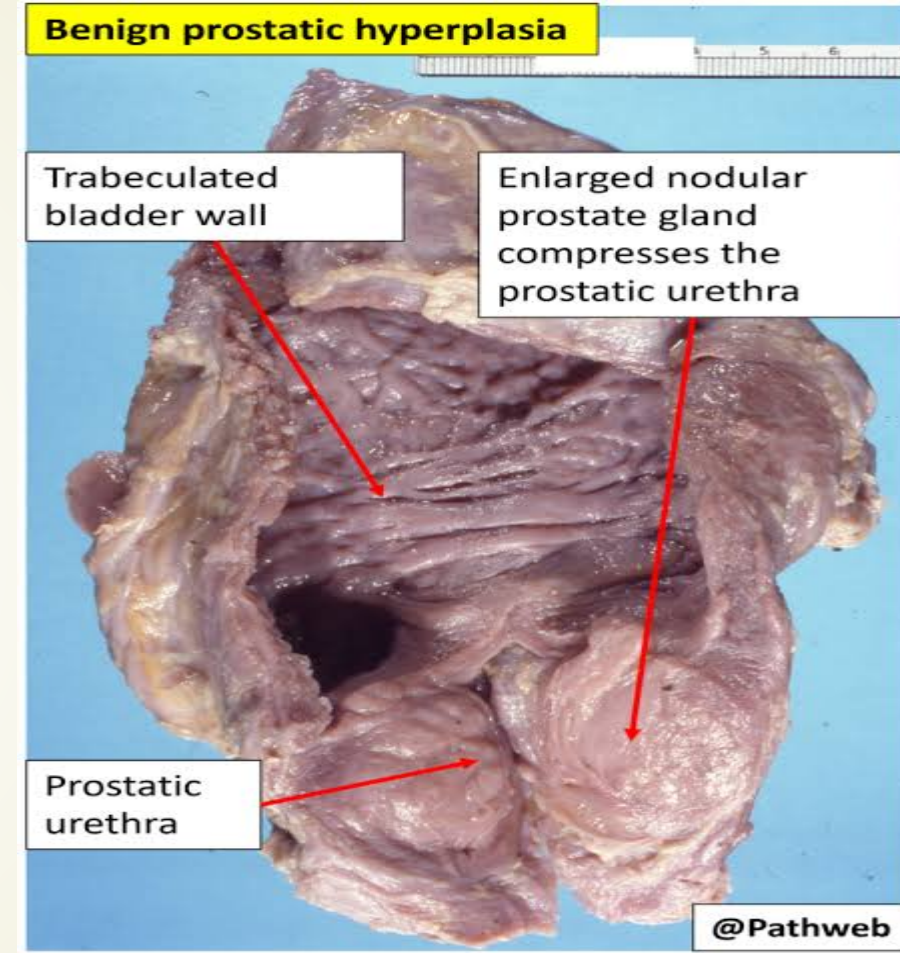
Cells forming both the endometrial glands and the stroma have increased in .number



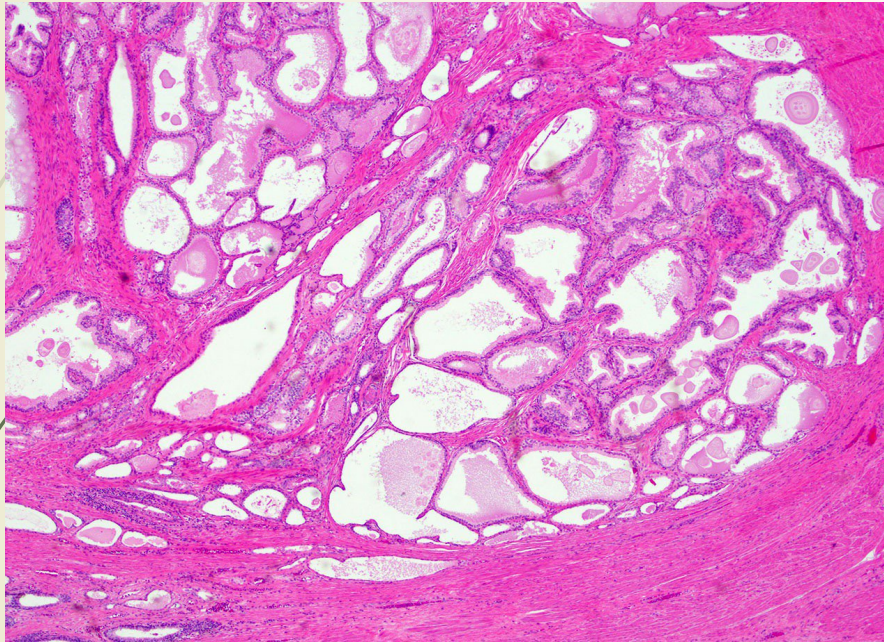
Hyperplasia



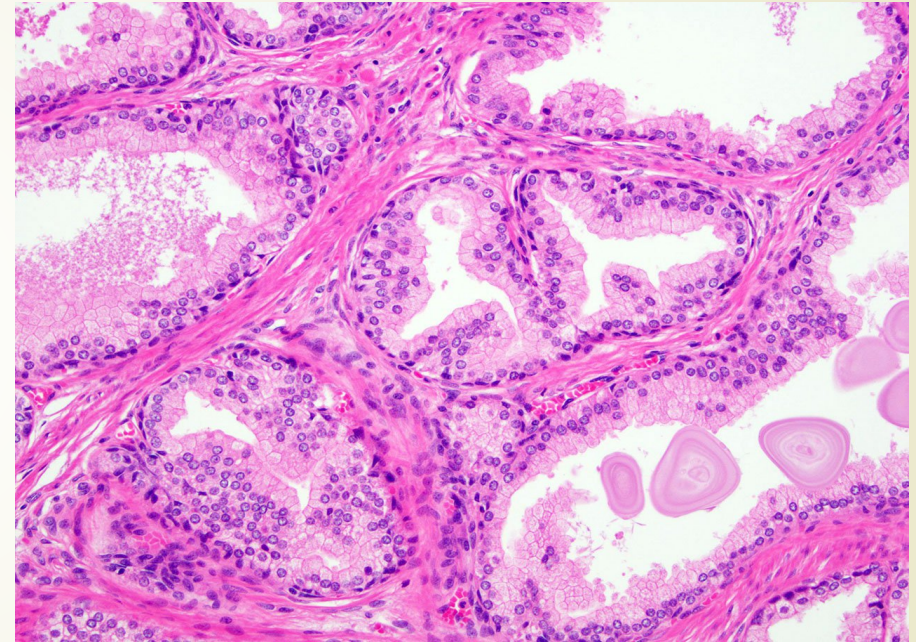
Benign prostatic hyperplasia (also known as nodular hyperplasia) causing obstructive urinary symptoms. The hyperplastic prostatic tissue is projecting into the bladder at the neck region



Benign prostatic hyperplasia



Well circumscribed glandular proliferation with variable sized glands with focal cystic dilatation and corpora amylacea.



benign prostatic glands with luminal infolding, lined by an inner layer of cuboidal to columnar secretory cells, with pale cytoplasm, regular nuclei



Atrophy

- Shrinkage in the size of the cell by the loss of cell substance is known as atrophy.
 - When a sufficient number of cells are involved, the entire tissue or organ diminishes in size, becoming atrophic.
- 

Atrophy as seen in the brain.

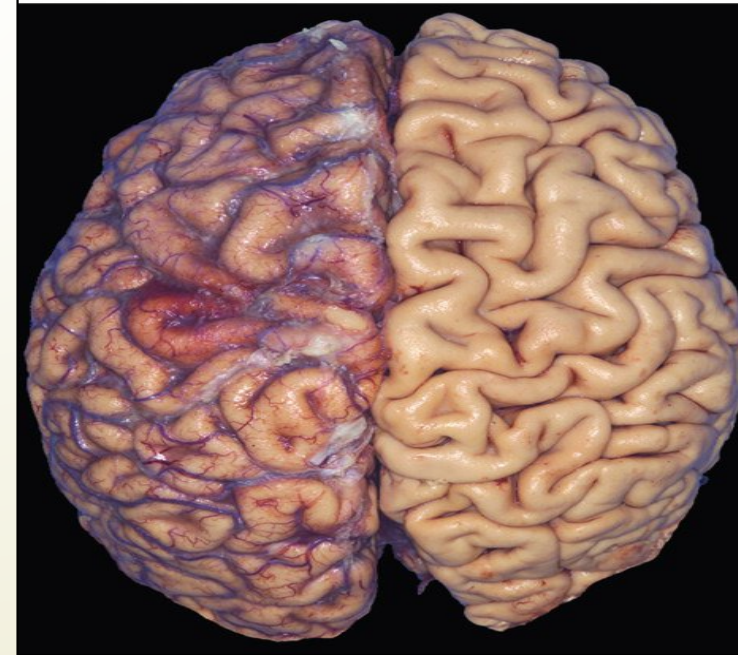
A- Normal brain of a young adult.

B- Atrophy of the brain in an 82-year-old man with atherosclerotic disease.

Atrophy of the brain is due to aging and reduced blood supply. Note that loss of brain substance narrows the gyri and widens the sulci.



B



A

Atrophy



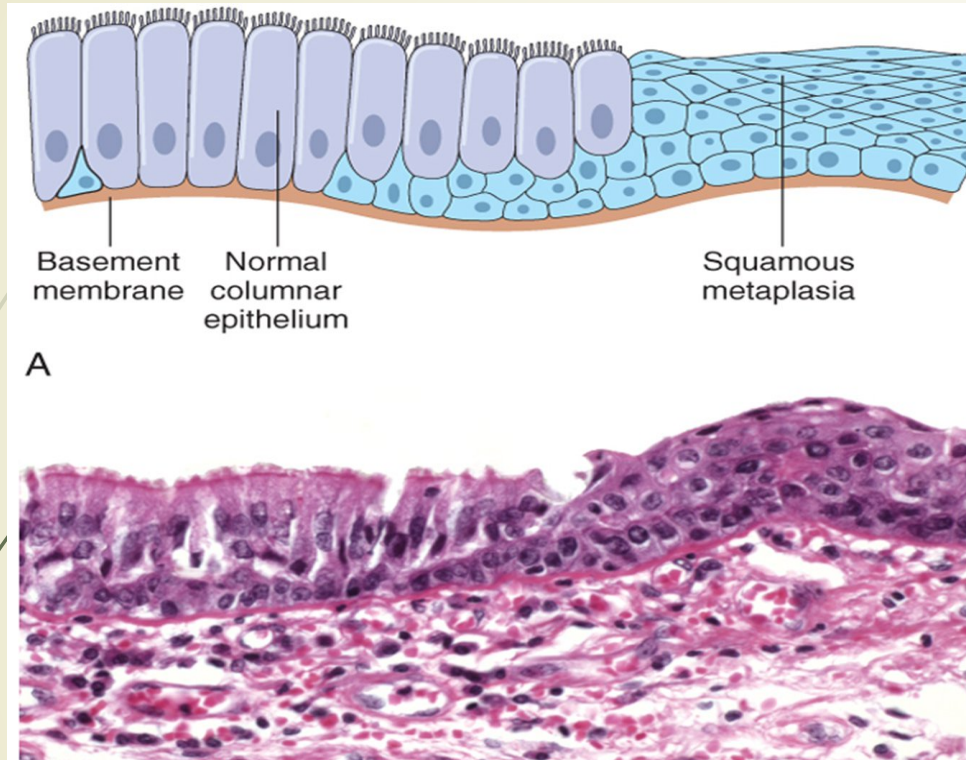
Muscle atrophy is the deterioration of muscle tissue due to extended periods of minimal use. In other words, muscles begin to waste away if they are not exercised consistently.



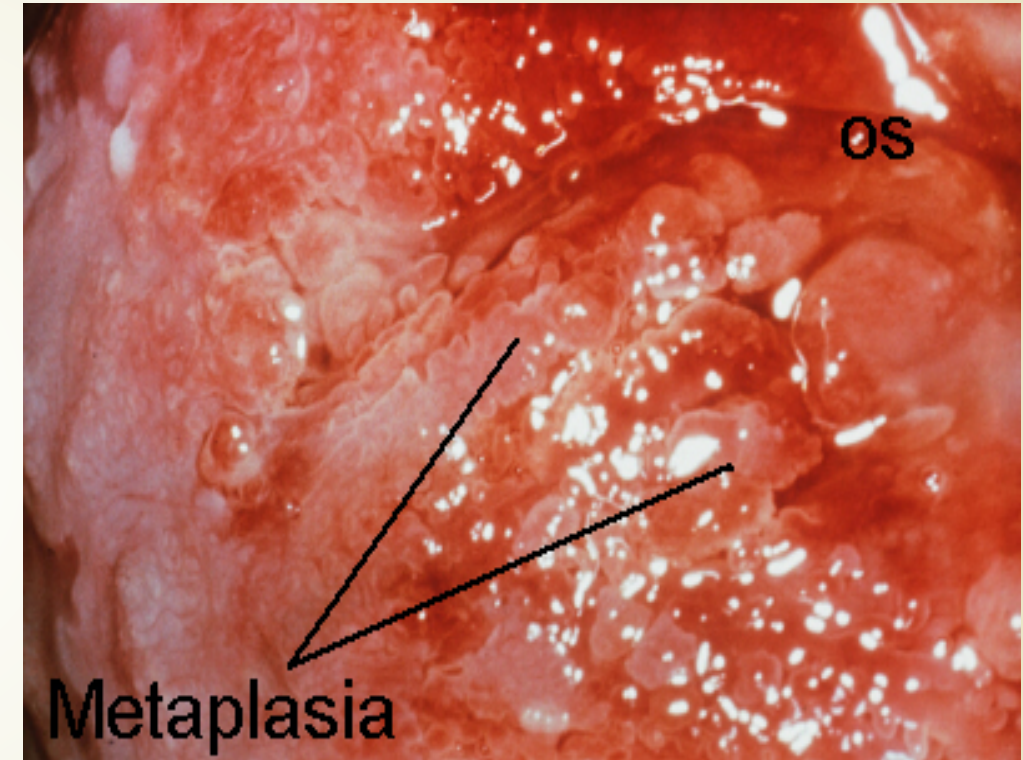
Metaplasia

- Metaplasia is a reversible change in which one adult cell type is replaced by another adult cell type, usually in response to chronic stress or irritation, such as exposure to smoke or stomach acid.
- While adaptive and reversible if the stressor is removed, persistent metaplasia can become dysplastic, a precancerous condition that, if left untreated, can progress to cancer.

Metaplasia



Metaplasia of normal columnar (left) to squamous epithelium (right) in a bronchus, shown (above) schematically and (down) histologically.

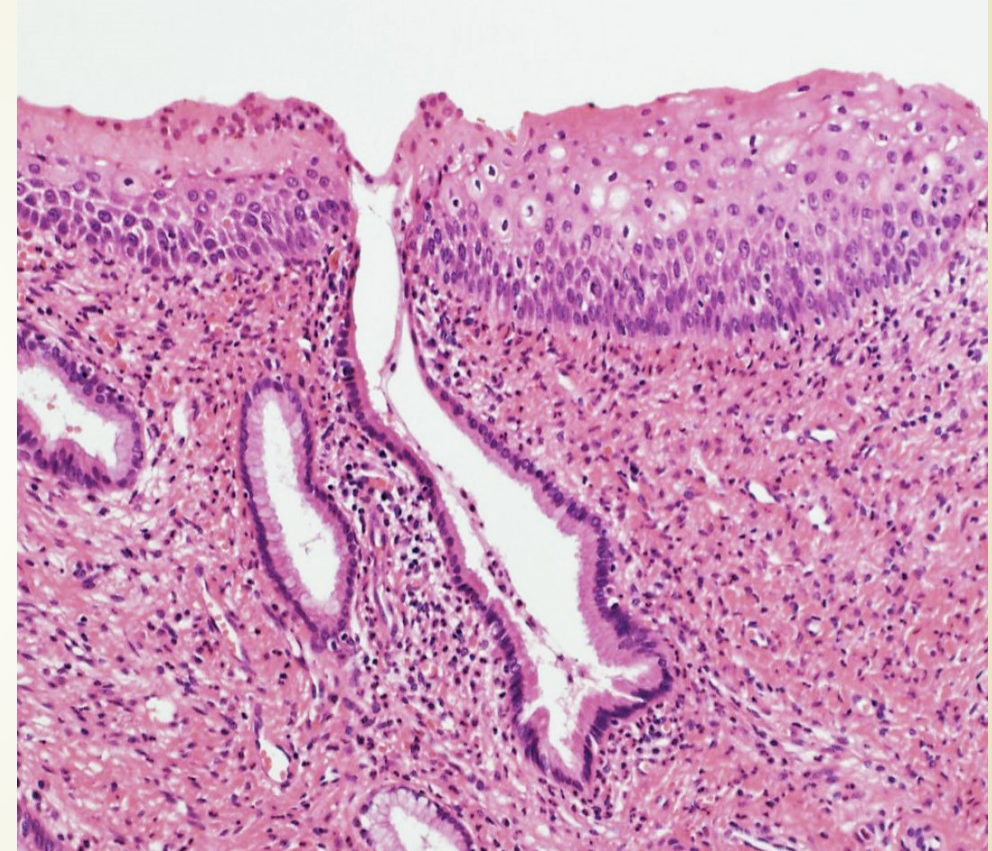


Mature squamous metaplasia

Cervix Metaplasia (Squamous metaplasia)



,Colposcopic appearance of transformation zone characterized by whitish and rather translucent metaplastic epithelium. Stratified squamous epithelium replacing preexisting mucin producing columnar epithelium



Opening of endocervical gland through metaplastic squamous metaplasia can be identified.



:Intracellular accumulations

- Cells may accumulate abnormal amounts of various substances; these may be harmless or associated with injury.
 - The locations of these substances are either cytoplasmic within organelles (typically lysosomes), or in the nucleus.
- 

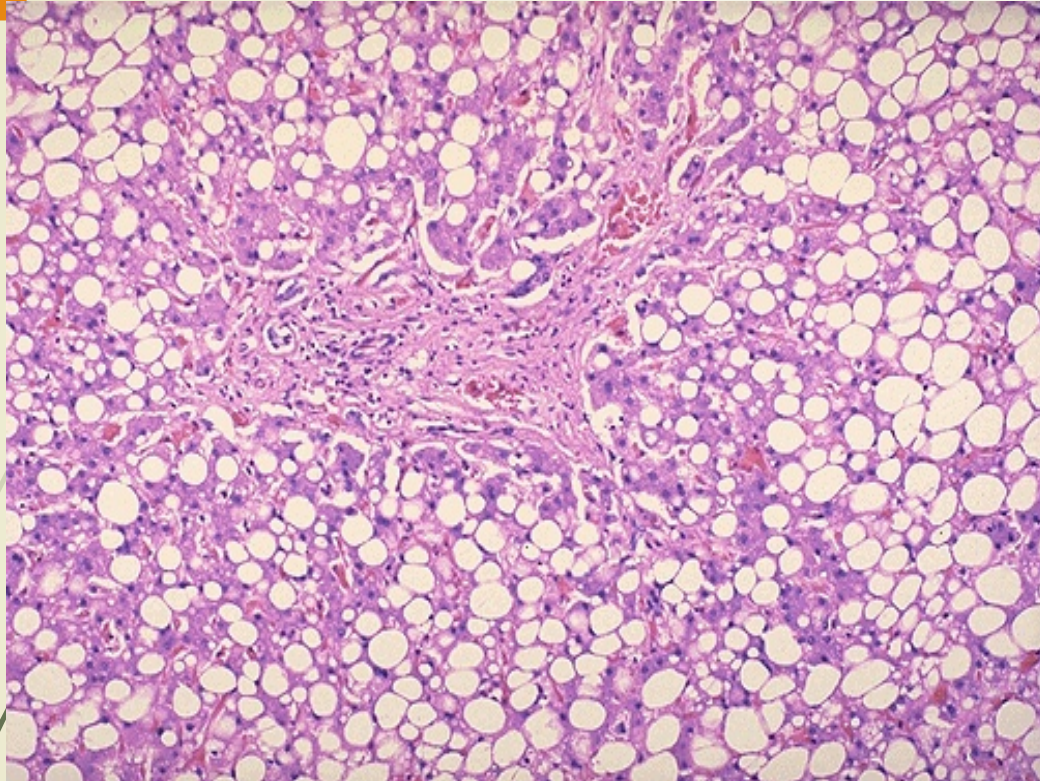
Fatty Change (Steatosis)

Severe fatty change liver
- abnormal accumulation
of triglycerides within
parenchymal cells.

- In the liver mild fatty
change shows no gross
changes with progressive
accumulation, the organ
enlarges and become
increasingly yellow, soft
and greasy to touch.

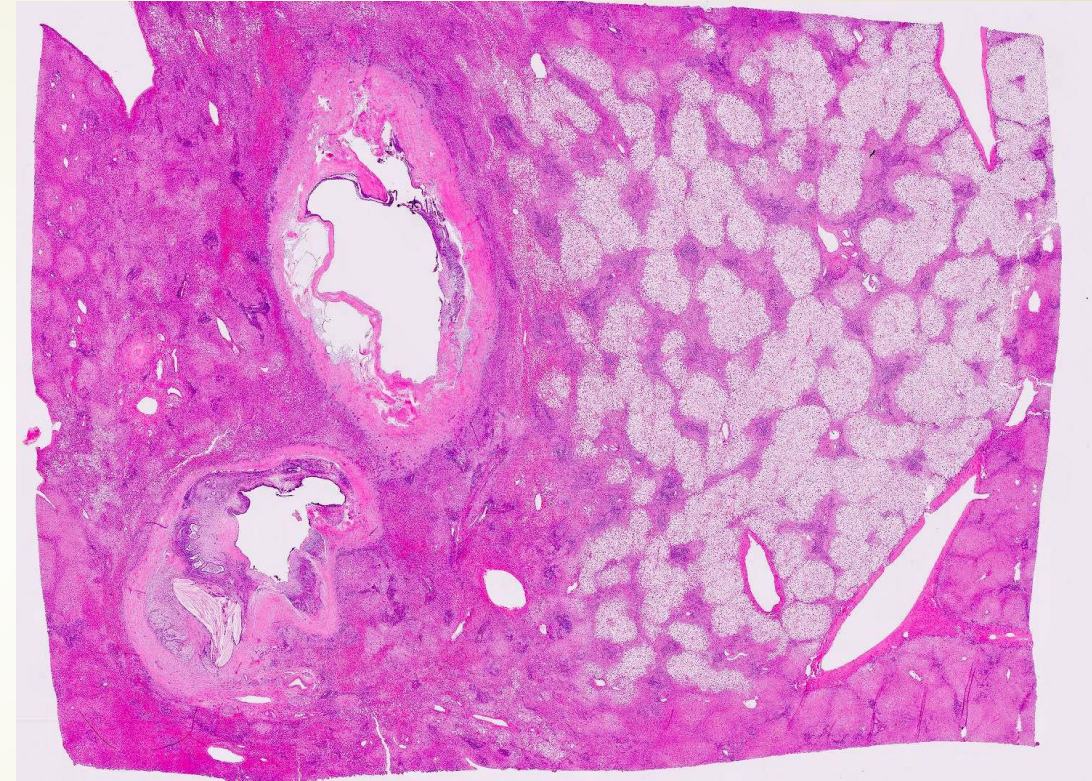


Fatty change liver



In the early stages: small fat vacuoles around the nucleus.

with progression these coalesce into a large clear space that displaces the nucleus to the periphery (macrovesicular steatosis).



diffuse steatosis on the right side of the image, compared to relatively benign liver parenchyma on the left

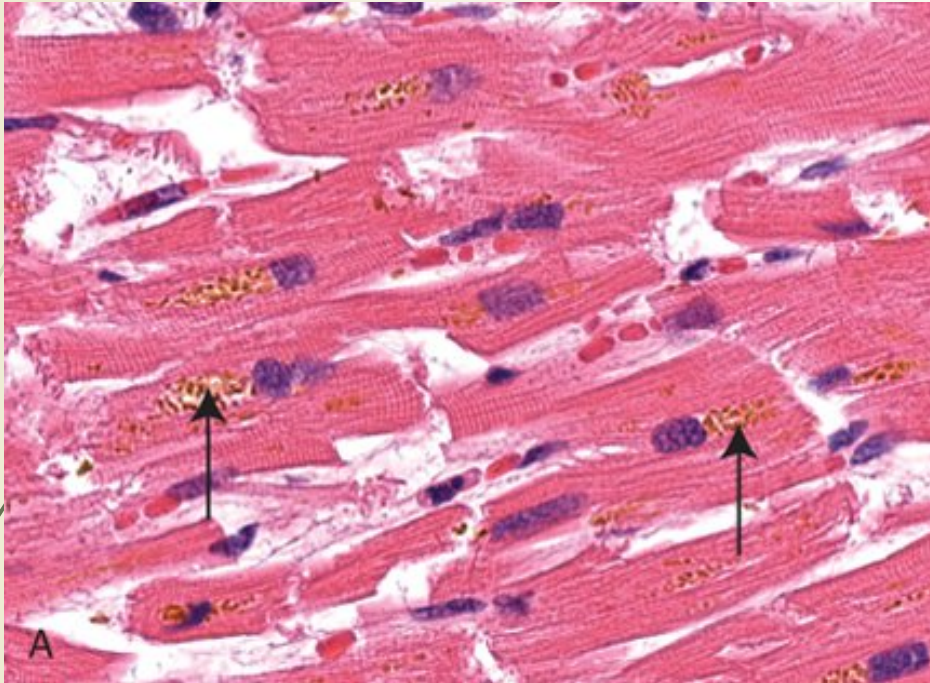


Pigments

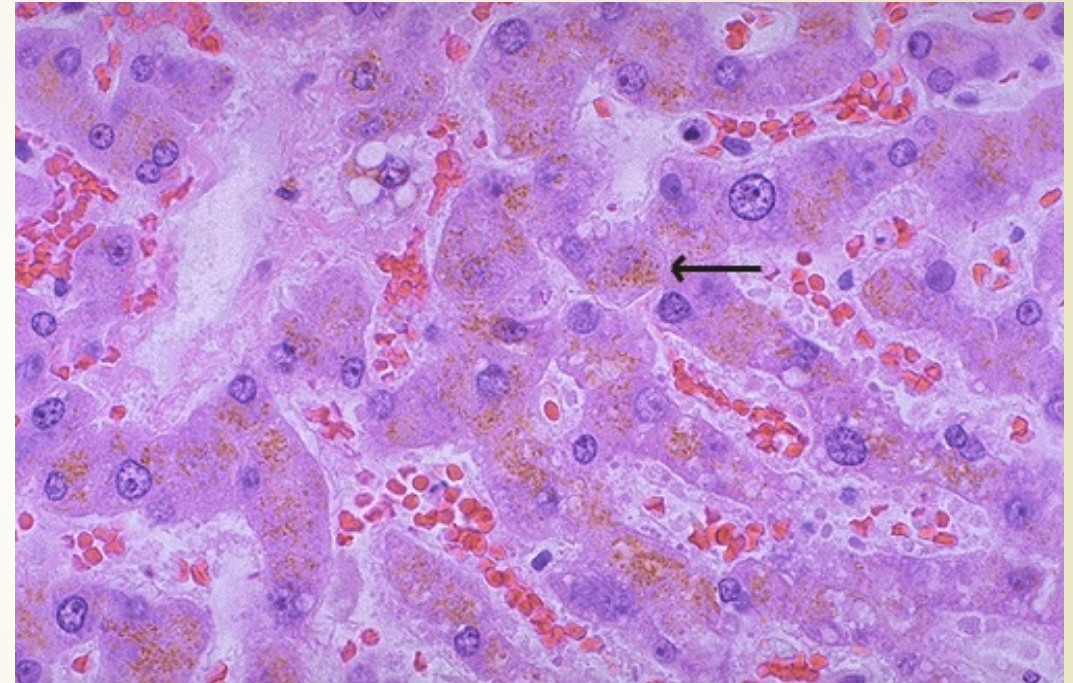
- Pigments are colored substances.
 - Exogenous, coming from outside the body, or
 - Endogenous, synthesized within the body itself
- 

Lipofuscin

It is an insoluble brownish-yellow granular intracellular material represent complex of protein and lipid.



- Lipofuscin granules in a cardiac myocytes
- Brownish-yellow granular intracellular material (deposits indicated by arrows).



Lipofuscin pigment in the cytoplasm of hepatocytes, H&E stain.



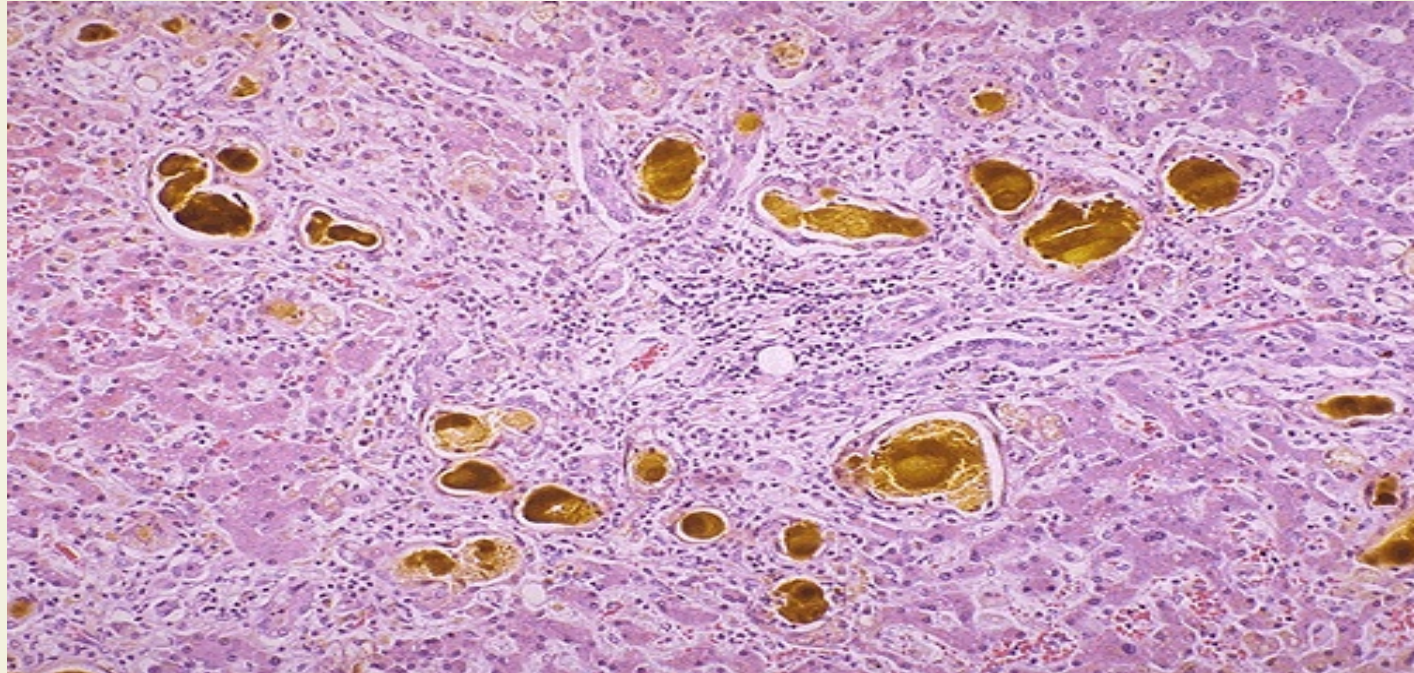
Bilirubin

- This is a normal major pigment of bile.
 - is derived from Hb (but unlike hemosiderin contains no iron).
 - Jaundice results from excess bilirubin pigment that is distributed throughout all tissues and body fluids.
- 

bilirubin accumulation
in skin and sclera



Jaundice



The yellow-green globular material seen within small bile ductules in the liver is bilirubin pigment. This is hepatic cholestasis. A problem with excretion of bile, or an obstruction to flow of bile, could cause this appearance. This patient would exhibit jaundice.



Hemosiderin

- This iron-containing pigment consists of aggregates of ferritin.
- It appears in tissues as golden brown amorphous aggregates.
- Identified by its staining reaction (blue color) with Prussian blue dye.

Hemosiderosis:

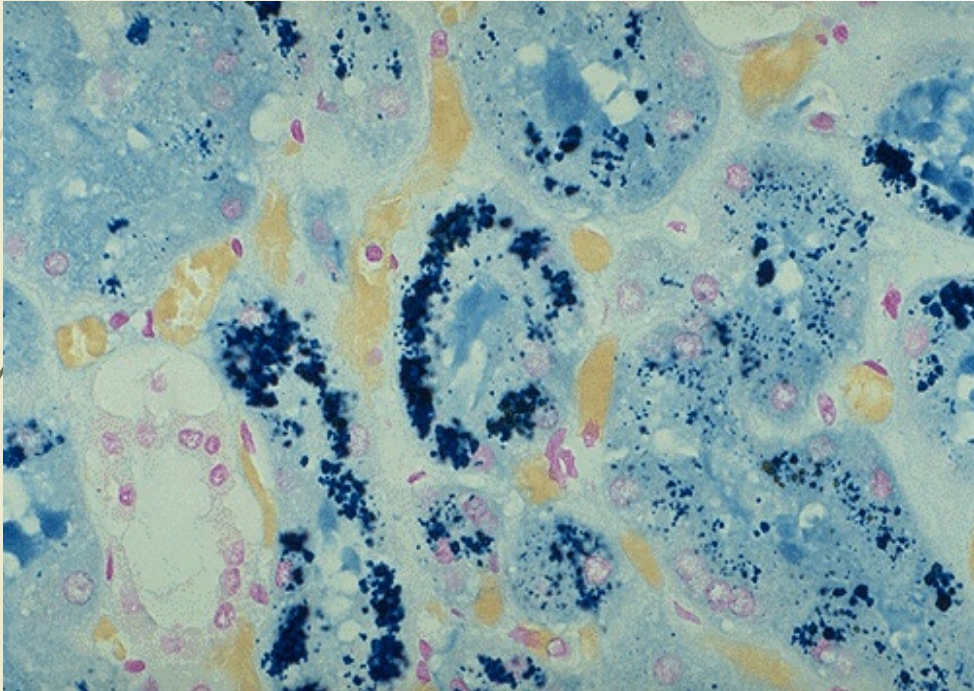
- Is defined by accumulation of hemosiderin, primarily within tissue macrophages.
- Not associated tissue or organ damage.

Hemochromatosis:

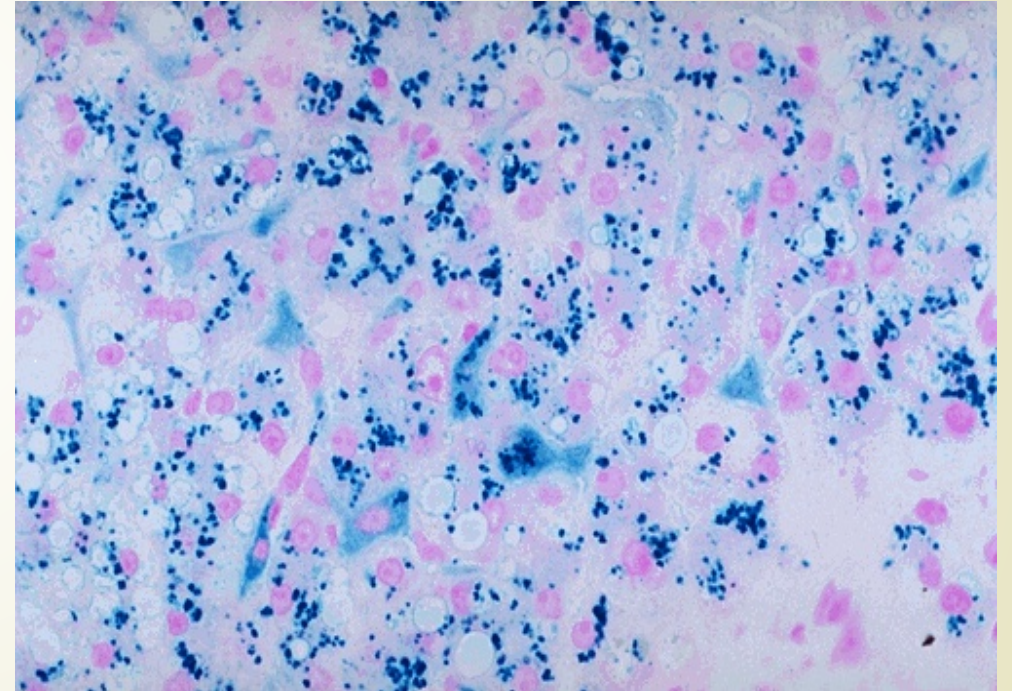
- Is more extensive accumulation of hemosiderin, often within parenchymal cells.
- associated with tissue damage, scarring, and organ dysfunction.

Hemosiderin granules

Iron deposits shown by a special staining process called the Prussian blue reaction.



These renal tubules contain large amounts of hemosiderin, as demonstrated by the Prussian blue iron stain. This patient had chronic hematuria.



Hemosiderin, liver, iron stain.

Pathologic calcification

- Pathologic calcification is a common process in a wide variety of disease states.
- it implies the abnormal deposition of calcium salts.

dystrophic calcification:

- When the deposition occurs in dead or dying tissues.

metastatic calcification:

- the deposition of calcium salts in normal tissues.

Dystrophic Calcification

Dystrophic calcification is encountered in:

1. Areas of necrosis (of any type as coagulative, caseous, etc.)
 2. Advanced atherosclerosis (in the aorta and coronaries)
 3. Aging or damaged heart valves resulting in severely impaired valve motion.
- Dystrophic calcification of the aortic valves is an important cause of aortic stenosis in the elderly.

Dystrophic calcification

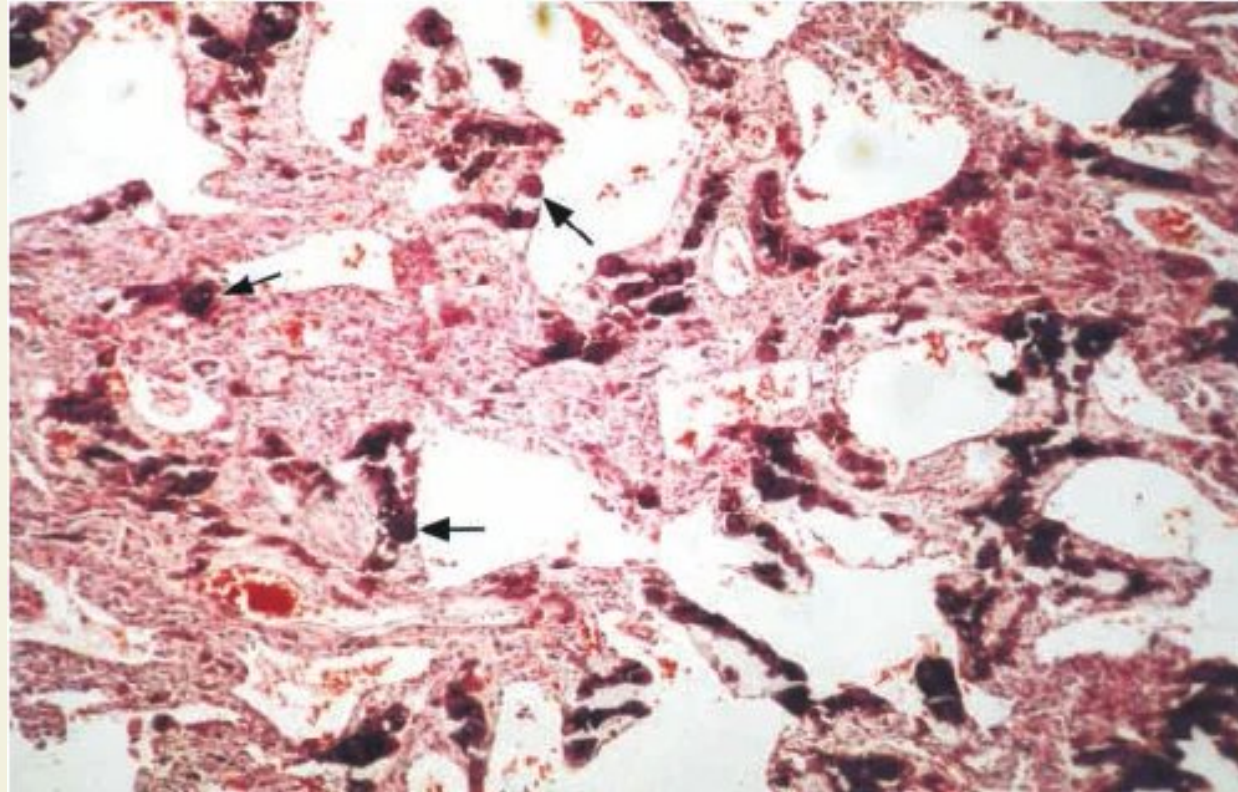


A view looking down onto the unopened aortic valve in a heart with calcific aortic stenosis.

The semilunar cusps are thickened and fibrotic. Behind each cusp are large, irregular masses of dystrophic calcification that will prevent normal opening of the cusps.

Metastatic calcification

Metastatic calcification can occur widely throughout the body but principally affects the interstitial tissues of the vessels, kidneys, lungs, and gastric mucosa



Histopathology of pulmonary metastatic calcification in a patient with CRF. Note that the calcified lesions (arrows) assume various shapes and are primarily located in the thickened fibroproliferative interstitium of the lungs

